Advance Praise for

Inflation in Emerging and Developing Economies

Many emerging market economies experienced a remarkable decline in inflation rates over the last two decades, after years of seemingly intractable high inflation. This is the first book-length analysis of this remarkable achievement, asking how it happened, what it tells us about best policy frameworks, and whether it will endure. At a time when global financial conditions pose a challenge to emerging-market currencies and monetary policies, this book is an essential guide to the road ahead. All students of the global economy will want to read it carefully.

Maurice Obstfeld, Economic Counsellor and Director of Research, International Monetary Fund

A remarkable resource for anyone interested inflation in the modern world, clear and easy to follow. This book is an order of magnitude more comprehensive than anything else out there, not only in its country coverage, but in its exploration of all the major issues and debates surrounding inflation. Curiously, most of the existing academic literature has focused on advanced economies – which are also thoroughly covered here -- yet there is so much to be learned from the dramatic inflation decline in emerging markets and low income economies, including for design of advanced economy institutions. Any student, academic researcher or policy economist who wants to understand the big picture on world inflation, and when and where it might surprise in the future, will find this book fascinating.

Kenneth Rogoff, Thomas D. Cabot Professor of Public Policy and Professor of Economics, Harvard University

THE WORLD BANK
IBRD • IDA | WORLD BANK GROUP
Development Economics • Prospects

Inflation in Emerging and Developing Economies: Evolution, Drivers, and Policies

Jongrim Ha, M. Ayhan Kose, and Franziska Ohnsorge

www.worldbank.org/inflation

THE WORLD BANK
IBRD • IDA | WORLD BANK GROUP
Development Economics • Prospects

Why this book?

1 **Debate on inflation.** *Explores whether global inflation could make a comeback after a prolonged period of low inflation.*

2 **Inflation in EMDEs.** *Presents the first comprehensive analysis of the evolution and drivers of inflation in EMDEs whereas previous literature mostly focuses on advanced economies.*

3 **Cutting-edge methods.** *Examines EMDE inflation using cutting-edge empirical methods that have thus far mostly been employed in studies of inflation in advanced economies.*

4 **Database of global inflation.** *Introduces a truly global database of up to 175 economies (34 advanced economies, 141 EMDEs) for 1970-2018 for multiple measures of inflation and many country characteristics.*

** EMDEs = Emerging Market and Developing Economies*

Four Questions

1 **How has inflation evolved in EMDEs?** *Declined over time thanks to the sharp fall in global inflation, supported by cyclical and structural developments. If momentum for these developments wanes, maintaining low inflation may become as difficult as achieving it.*

2 **What have been the global and domestic drivers of inflation?** *A wide range of shocks… A global inflation cycle has emerged and inflation synchronization has strengthened among EMDEs over time. Global shocks have become more important over time but domestic shocks remain the main source of national inflation variation.*

3 **How well-anchored are EMDE inflation expectations?** *Despite becoming better-anchored since the 1990s, inflation expectations remain less well anchored in EMDEs than advanced economies. Stronger monetary frameworks and central bank independence are associated with better anchoring of expectations and lower exchange rate pass-through.*

4 **What are the main policy implications?** *Strong monetary policy frameworks, more central bank independence, robust exchange rate regimes and resilient fiscal policy frameworks are all necessary to achieve and maintain low and stable inflation.*

THE WORLD BANK
IBRD • IDA | WORLD BANK GROUP
Development Economics • Prospects

Four Questions

1 How has inflation evolved in EMDEs? *Declined over time thanks to the sharp fall in global inflation, supported by cyclical and structural developments. If momentum for these developments wanes, maintaining low inflation may become as difficult as achieving it.*

THE WORLD BANK
IBRD • IDA | WORLD BANK GROUP
Development Economics • Prospects

Global Database of Inflation and Country Characteristics

1 **Large number of countries and long time-series starting in 1970.** *A truly global database of up to 175 economies (34 advanced economies, 141 EMDEs) for 1970-2018.*

2 **Multiple measures of inflation from a wide range of sources.** *Six measures of inflation: four CPI measures (headline, core, food, and energy), PPI, and GDP deflator.*

3 **Comprehensive database of country characteristics.** *Global database of over 40 variables on macroeconomy, monetary policy and exchange rate regimes, and global integration between 1970-2018.*

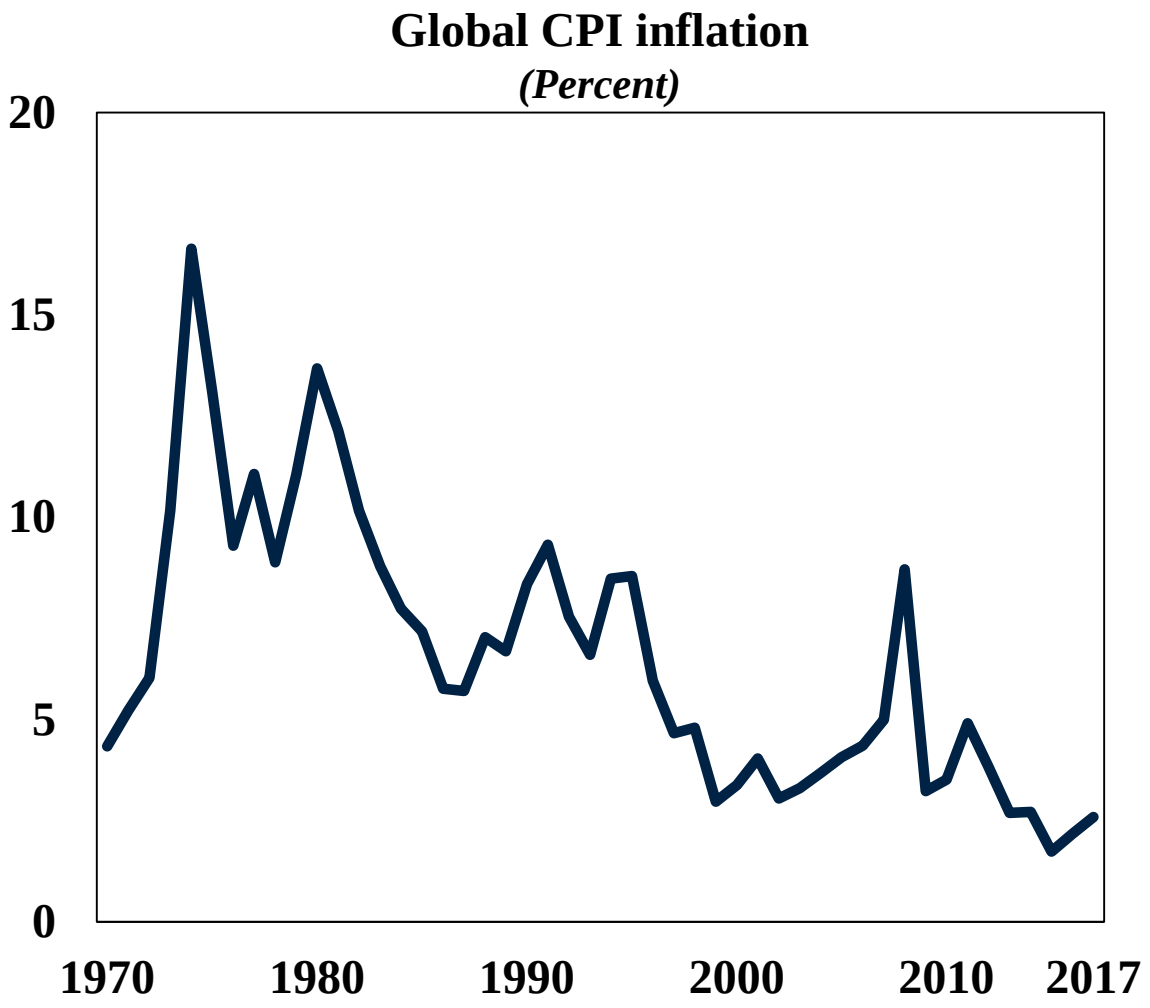
All charts/tables already posted on the web; Database coming soon…

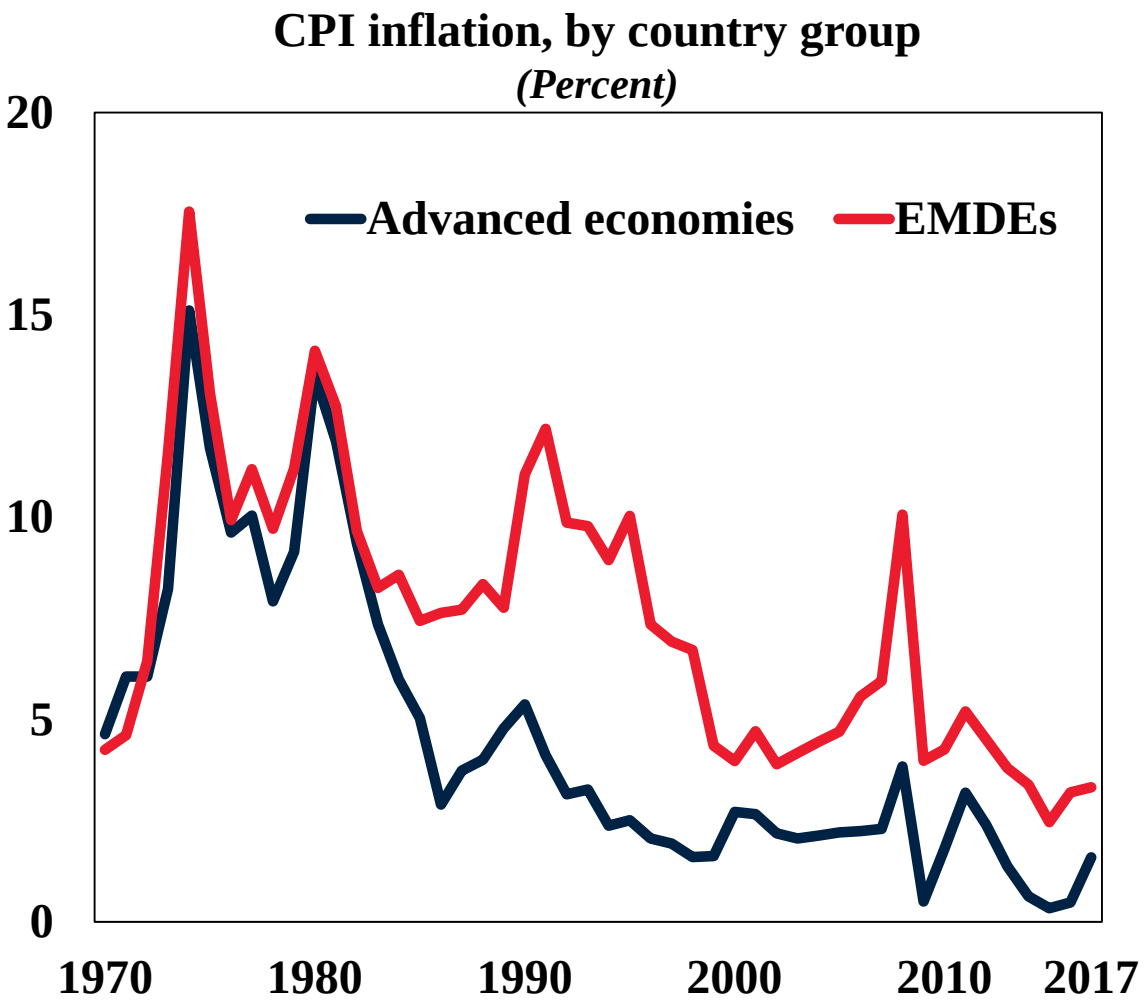
www.worldbank.org/inflation

THE WORLD BANK
IBRD • IDA | WORLD BANK GROUP
Development Economics • Prospects

Evolution of Inflation - 1

Synchronized Decline across Country Groups



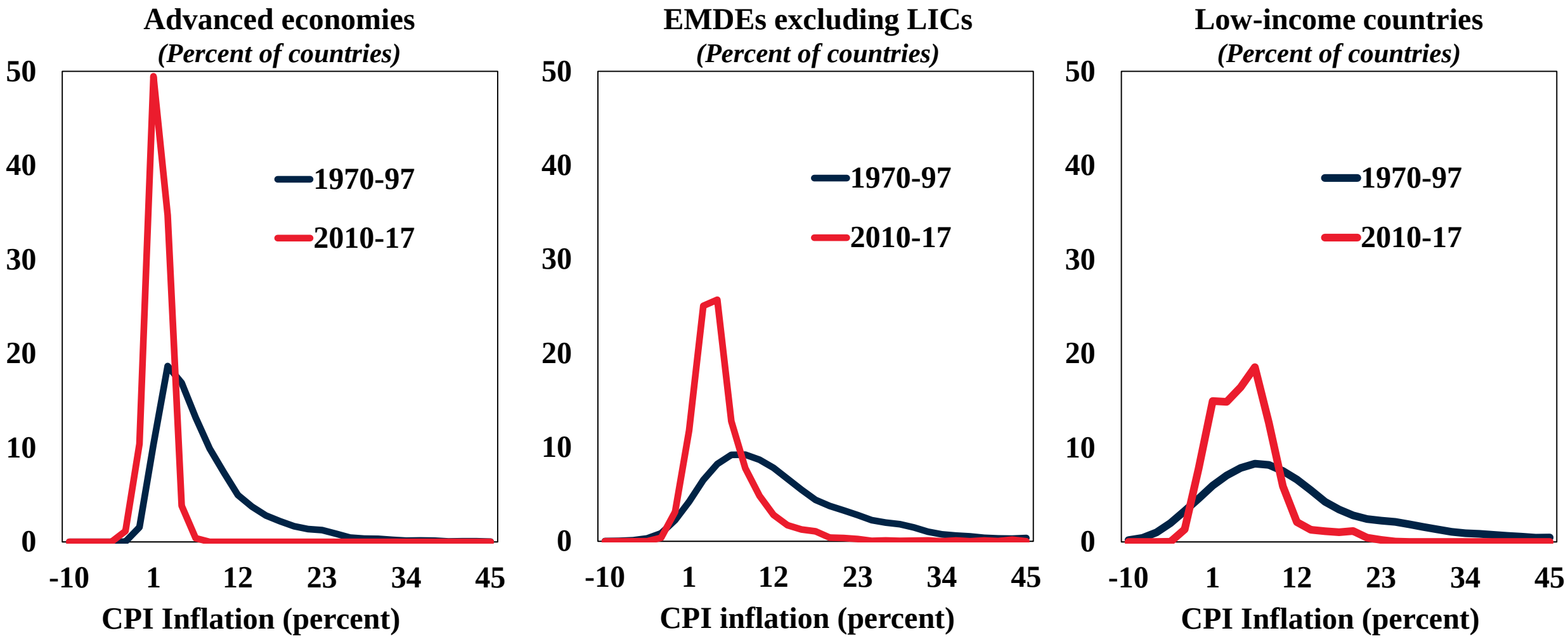


Source: World Bank.
Left Panel. Median headline CPI (consumer price index) inflation of 152 countries. Right Panel. Median headline CPI inflation, based on 29 advanced economies and 123 EMDEs.

THE WORLD BANK
IBRD • IDA | WORLD BANK GROUP
Development Economics • Prospects

Evolution of Inflation - 2

Broad-Based Decline within Country Groups

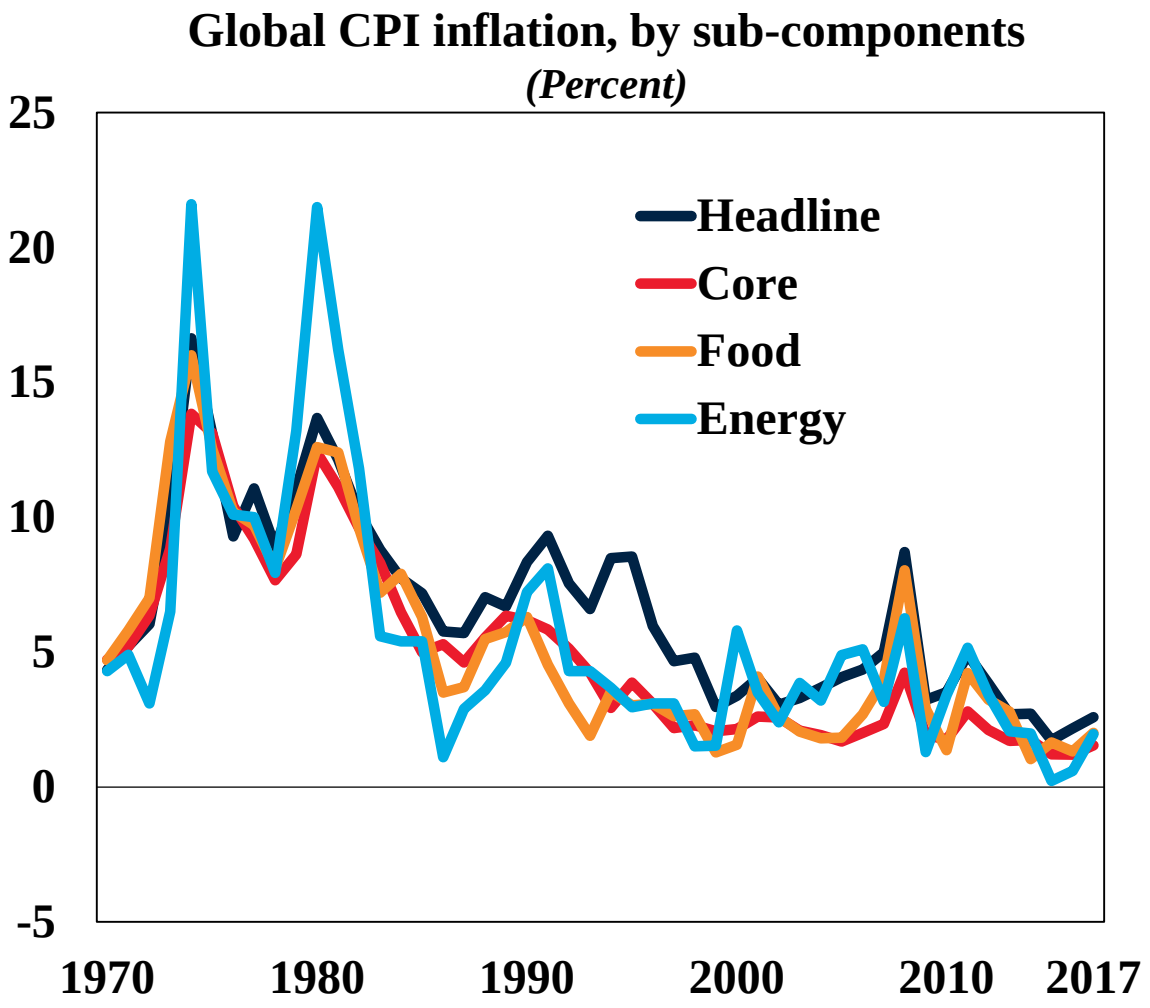


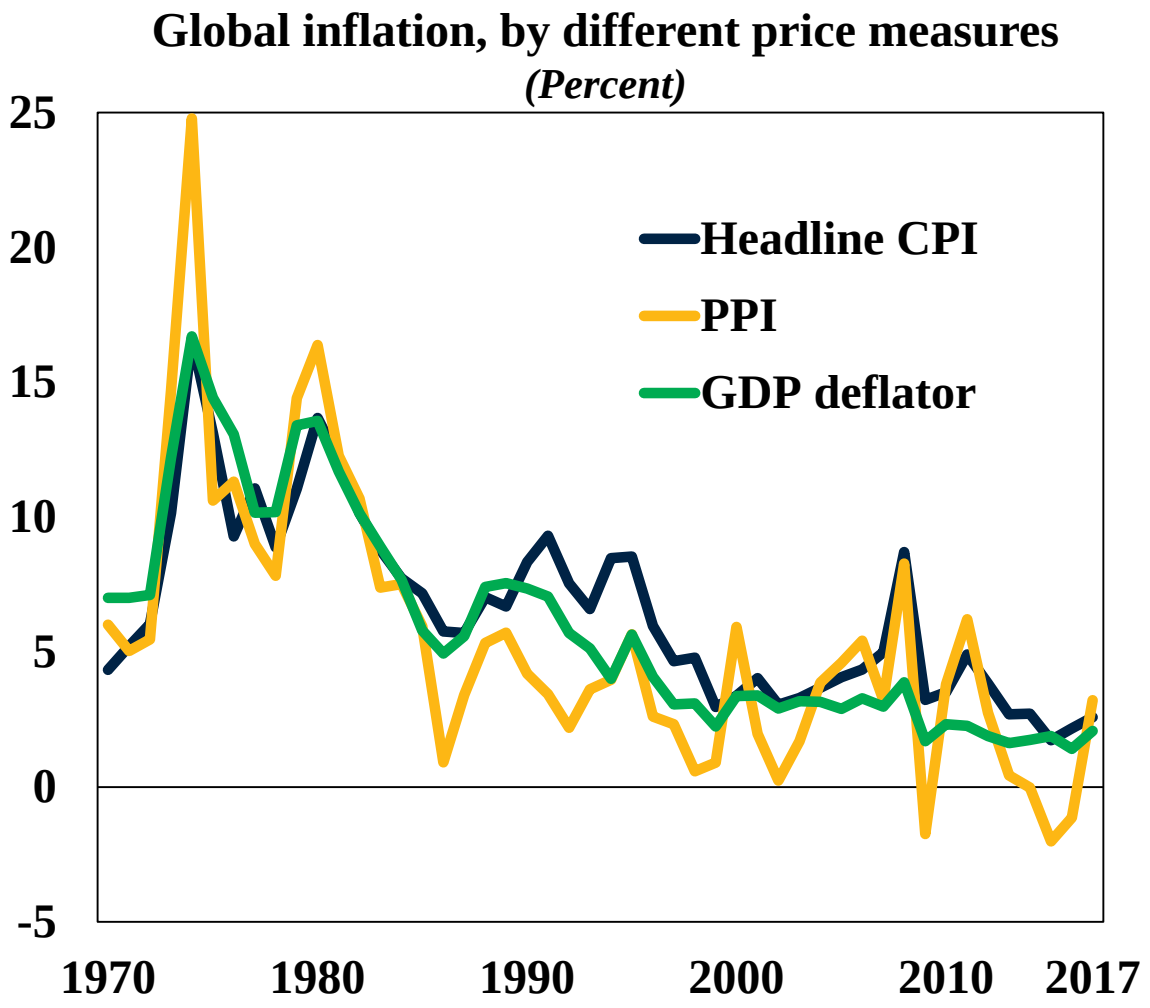
Source: World Bank.
Note: Kernel density estimation of the distribution of advanced economies (Left Panel) and EMDEs excluding low-income countries (LICs) (Center Panel), and LICs (Right Panel) based on the level of headline CPI inflation.

THE WORLD BANK
IBRD • IDA | WORLD BANK GROUP
Development Economics • Prospects

Evolution of Inflation - 3

Broad-Based Decline across Different Measures





Source: World Bank.
Note: Headline CPI (consumer price index) inflation is median of 152 countries. Left Panel. Core, food, and energy inflation is median of 47 countries. Right Panel. Inflation measured by PPI (producer price index) and GDP deflator is median of 39 countries, where data on PPI are consistently available over 1970-2017.

THE WORLD BANK
IBRD • IDA | WORLD BANK GROUP
Development Economics • Prospects

Factors Associated with Disinflation during 1970-2017

1 Globalization

- *Greater trade and global value chain integration can improve competition and productivity growth.*
- *Greater financial integration can increase central banks' anti-inflation bias.*

2 Better policy frameworks

- *Move to credible monetary policy frameworks and exchange rate regimes can help anchor inflation expectations.*
- *Greater central bank independence and transparency can improve anchoring of inflation expectations.*
- *Better fiscal frameworks can bolster credibility of monetary policy.*

3 Other structural factors

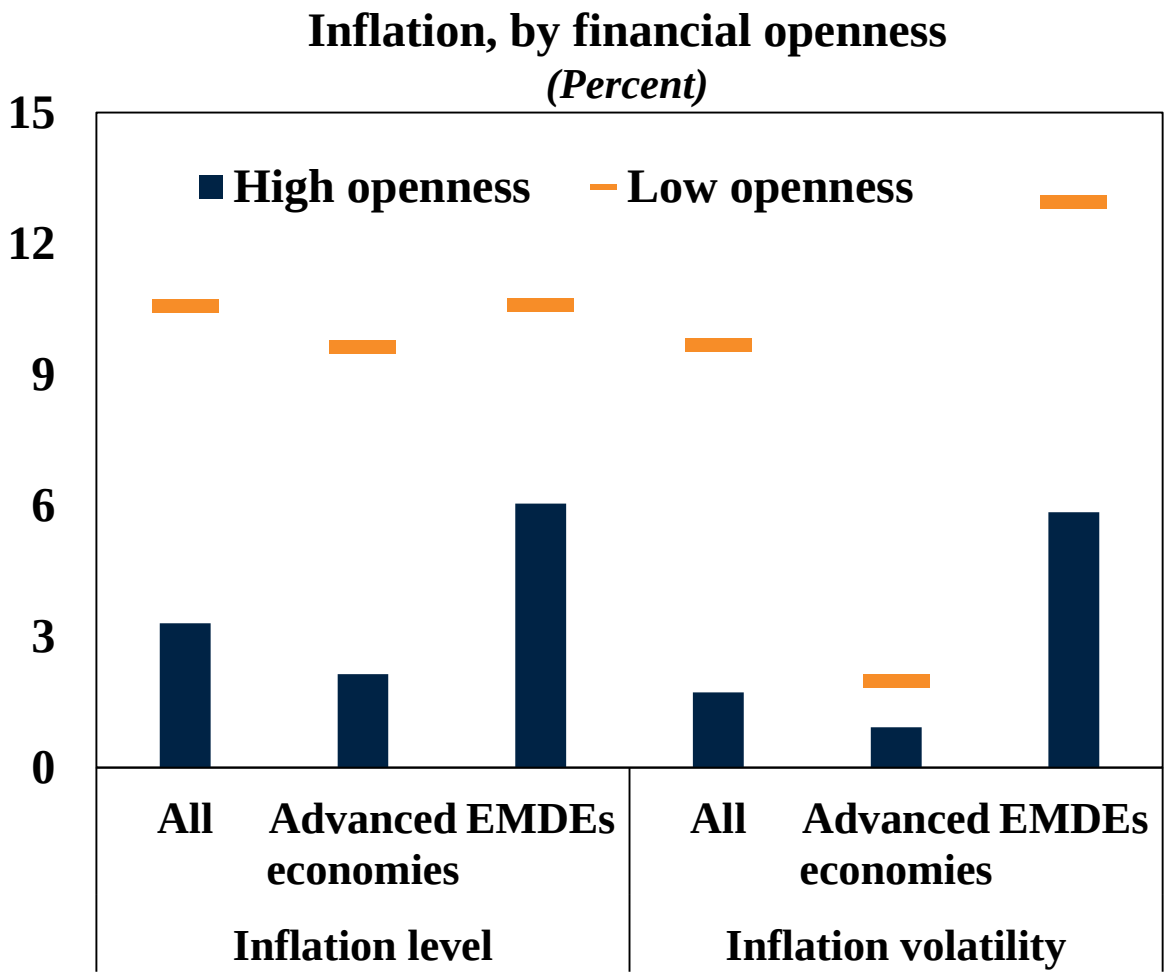
- *More flexible product and labor markets can increase competition and reduce wage rigidities.*
- *Population aging can dampen domestic demand growth.*
- *Digitalization can promote competition and productivity growth.*

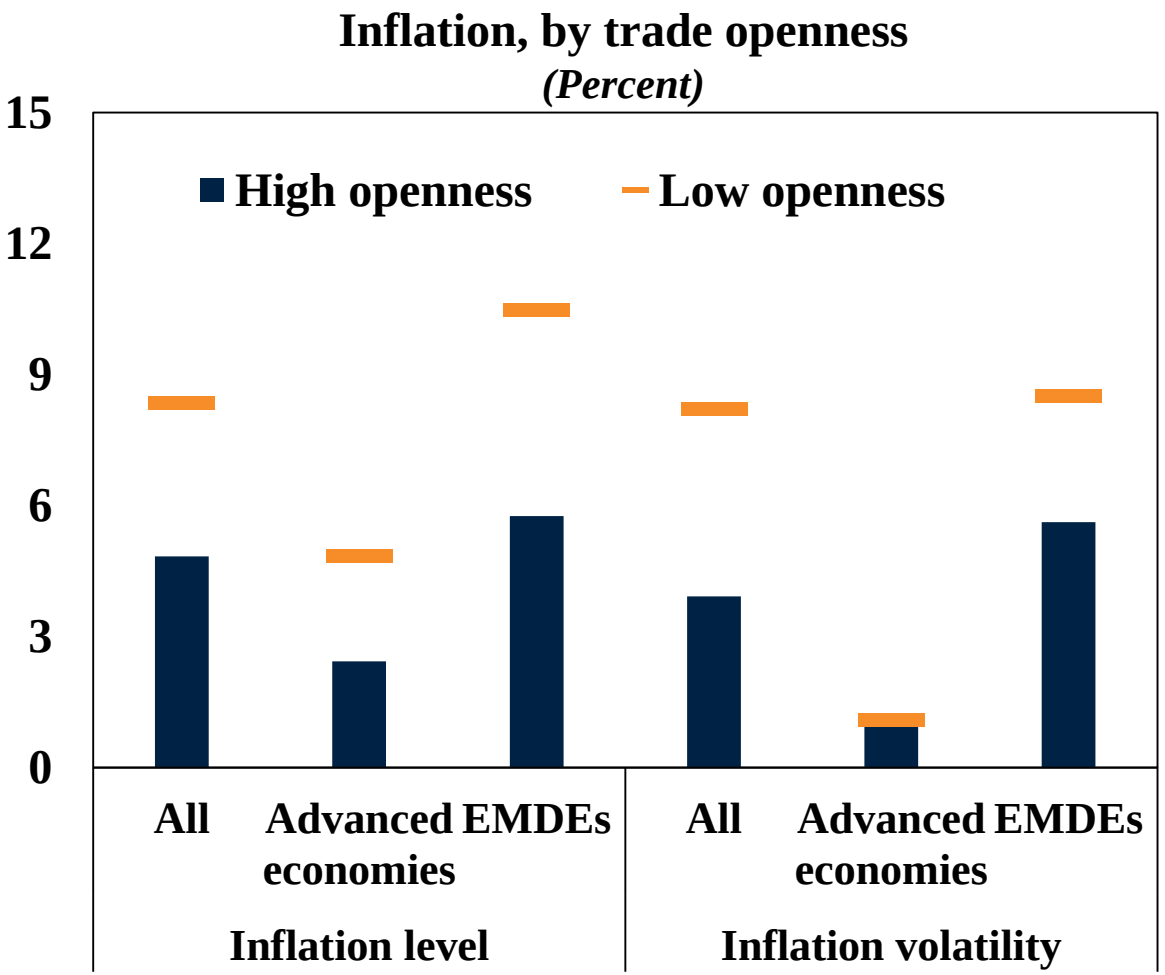
4 Multiple disinflationary shocks over past decade

- *Global financial crisis; Euro Area debt crisis*
- *2014-16 oil price plunge*

Factors Supporting Disinflation - 1

Globalization of Finance and Trade



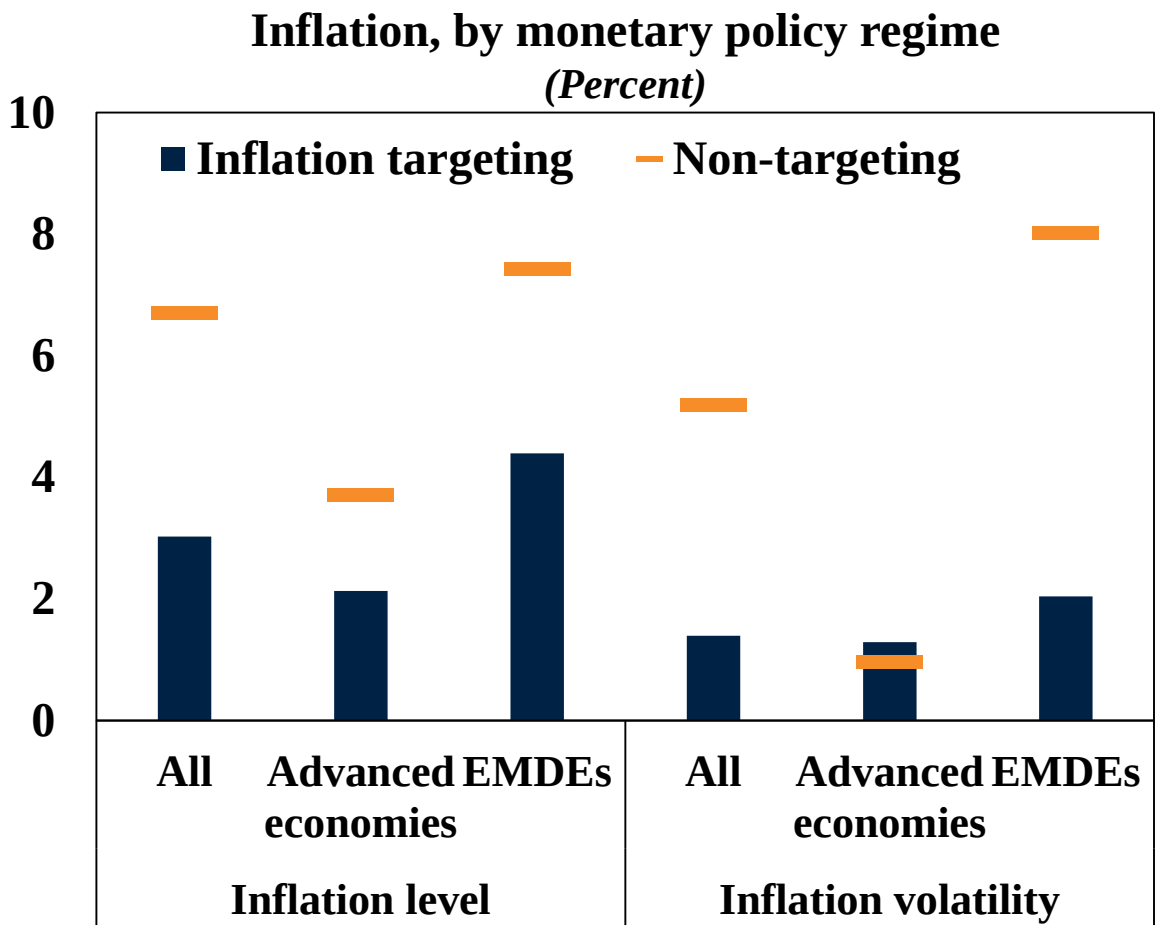


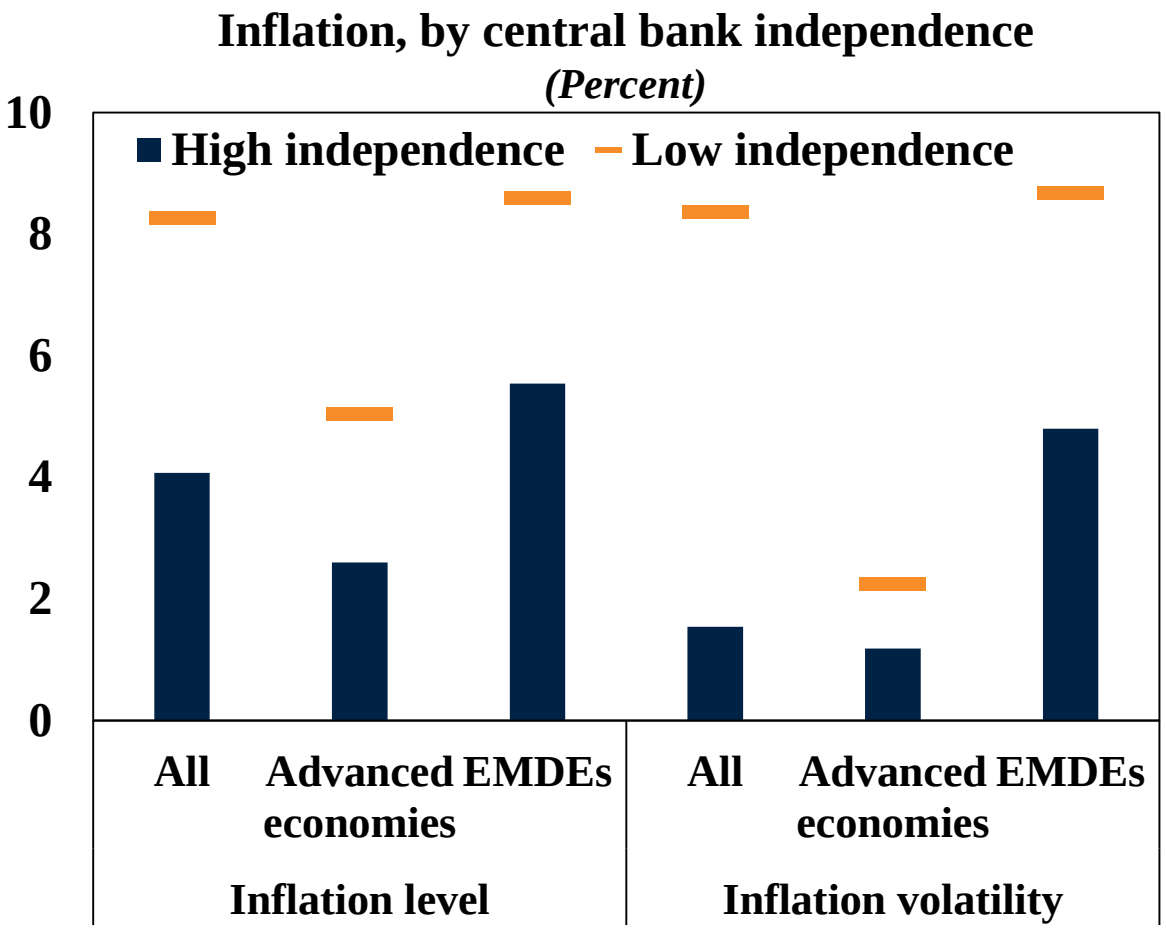
Sources: International Monetary Fund; Lane and Milesi-Ferretti (2007); World Bank.
Note: Blue bars indicate median inflation levels or inflation volatility (defined as standard deviation of inflation) in countries with financial openness (Left Panel) and trade openness (Right Panel) in the top quartile. Orange tickers indicate median inflation levels or inflation volatility in countries in the bottom quartile. Left Panel. Financial openness is measured as the sum of foreign assets and liabilities, as a share of GDP. Right Panel. Trade openness is measured as the sum of exports and imports as a percent of GDP.

THE WORLD BANK
IBRD • IDA | WORLD BANK GROUP
Development Economics • Prospects

Factors Supporting Disinflation - 2

Resilient Monetary Policy Frameworks; Central Bank Independence





Sources: Caceres, Carrière-Swallow, and Gruss (2016); Dincer and Eichengreen (2014); World Bank.
Left Panel. Blue bars show median inflation levels or inflation volatility in countries with inflation targeting monetary policy regimes during 1970-2017. Orange tickers indicate median inflation levels or inflation volatility in countries without inflation-targeting monetary policy regimes during the same period. Inflation targeting regimes are defined as in Caceres, Carrière-Swallow, and Gruss (2016) and the IMF Annual Report on Exchange Arrangements and Exchange Restrictions. Right Panel. Blue bars show median inflation levels or inflation volatility in countries with a score of the index of central bank independence in the top quartile of the sample. Orange tickers indicate median inflation levels or inflation volatility in countries in the bottom quartile. Central bank independence is measured by the index of central bank independence and transparency, taken from Dincer and Eichengreen (2014). The index ranges from 0 (least independent and transparent) to 15 (most independent and transparent).

THE WORLD BANK
IBRD • IDA | WORLD BANK GROUP
Development Economics • Prospects

Four Questions

1 **How has inflation evolved in EMDEs?** *Declined over time thanks to the sharp fall in global inflation, supported by cyclical and structural developments. If momentum for these developments wanes, maintaining low inflation may become as difficult as achieving it.*

2 **What have been the global and domestic drivers of inflation?** *A wide range of shocks… A global inflation cycle has emerged and inflation synchronization has strengthened among EMDEs over time. Global shocks have become more important over time but domestic shocks remain the main source of national inflation variation.*

Global and Group-Specific Drivers of Inflation

Dynamic Factor Model

- **Methodology:** Dynamic factor model with global and group-specific factors (Kose, Otrok, Prasad 2012)

$$\pi_t^{i,j} = \beta_{global}^{i,j} f_t^{global} + \beta_{group(j)}^{i,j} f_t^{group(j)} + \varepsilon_t^{i,j}$$

$$f_t^m = \varphi^m(L) f_{t-1}^m + \mu_t^m \quad for\ m = 1 \ldots (1+J)$$

$$\varepsilon_t^{i,j} = \varphi^{i,j}(L) \varepsilon_{t-1}^{i,j} + \nu_t^{i,j}$$

- Variance decompositions

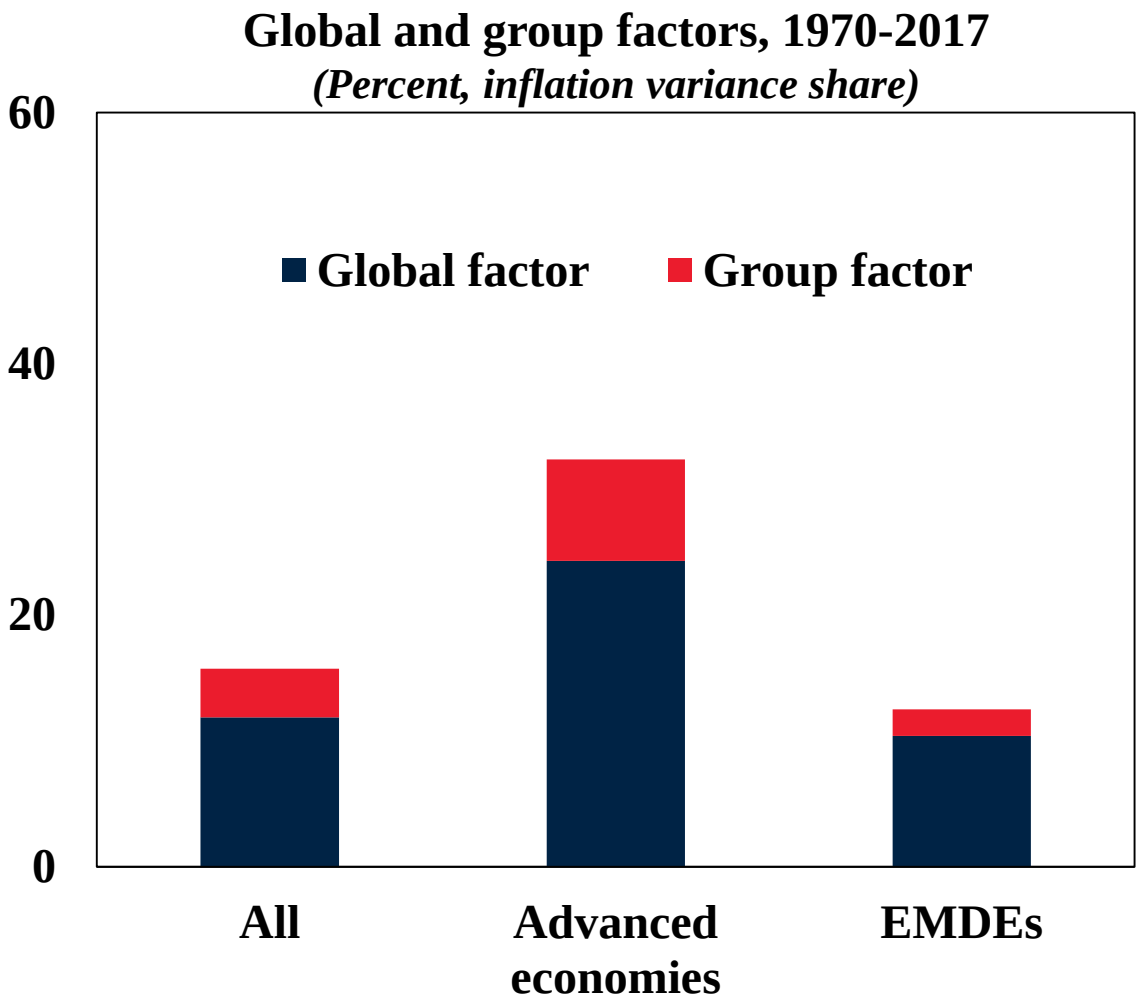
$$\mathrm{var}\left(\pi^{i,j} \mid f^{global}\right) = \frac{\left[\beta_{global}^{i,j}\right]^2 \mathrm{var}\left(f^{global}\right)}{\mathrm{var}\left(\pi^{i,j}\right)}$$

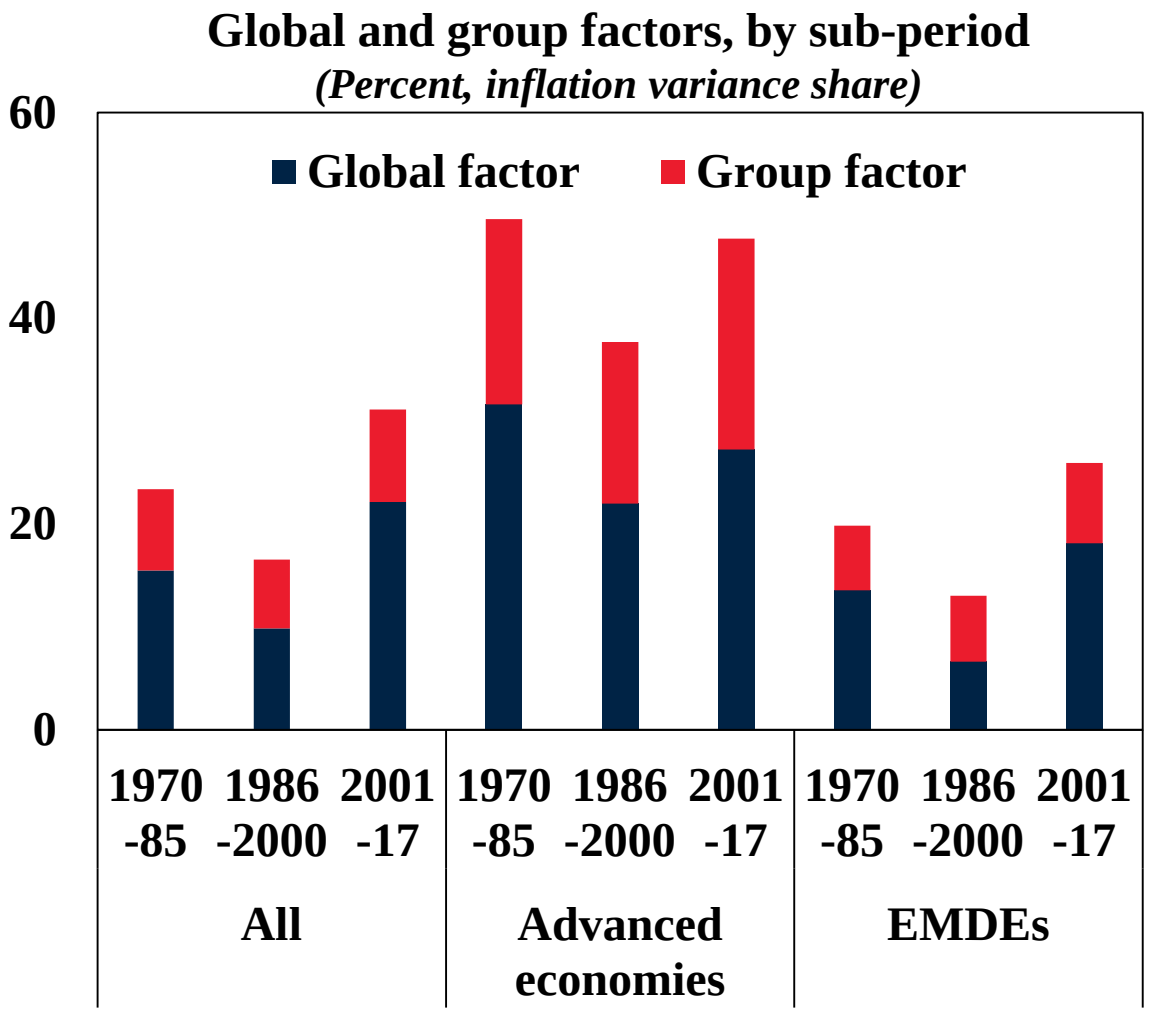
- **Database:** Annual data for 25 advanced economies and 74 EMDEs for 1970-2017

THE WORLD BANK
IBRD • IDA | WORLD BANK GROUP
Development Economics • Prospects

Inflation Synchronization Over Time

Global Factor Explains a Growing Share of Inflation Variation



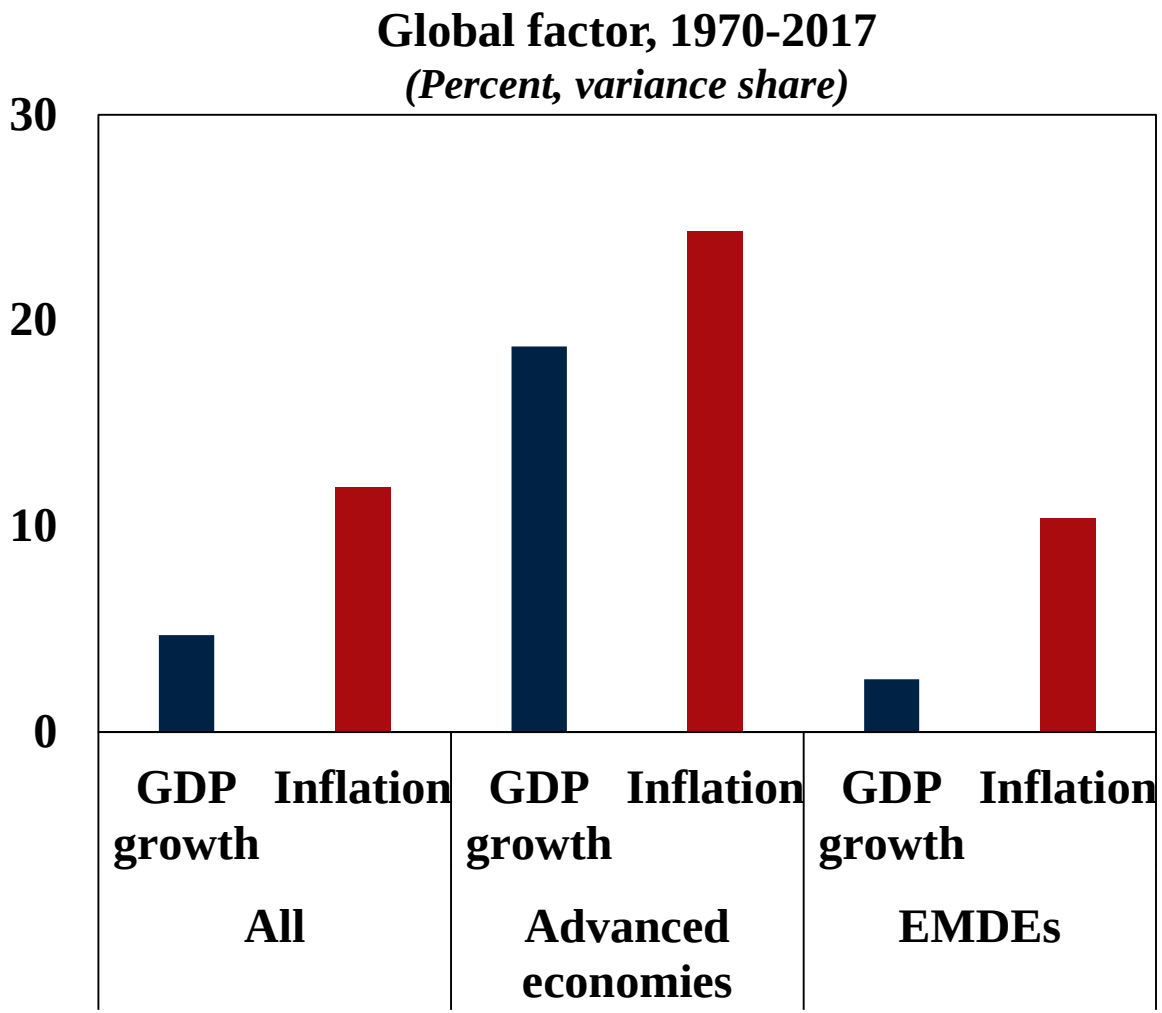


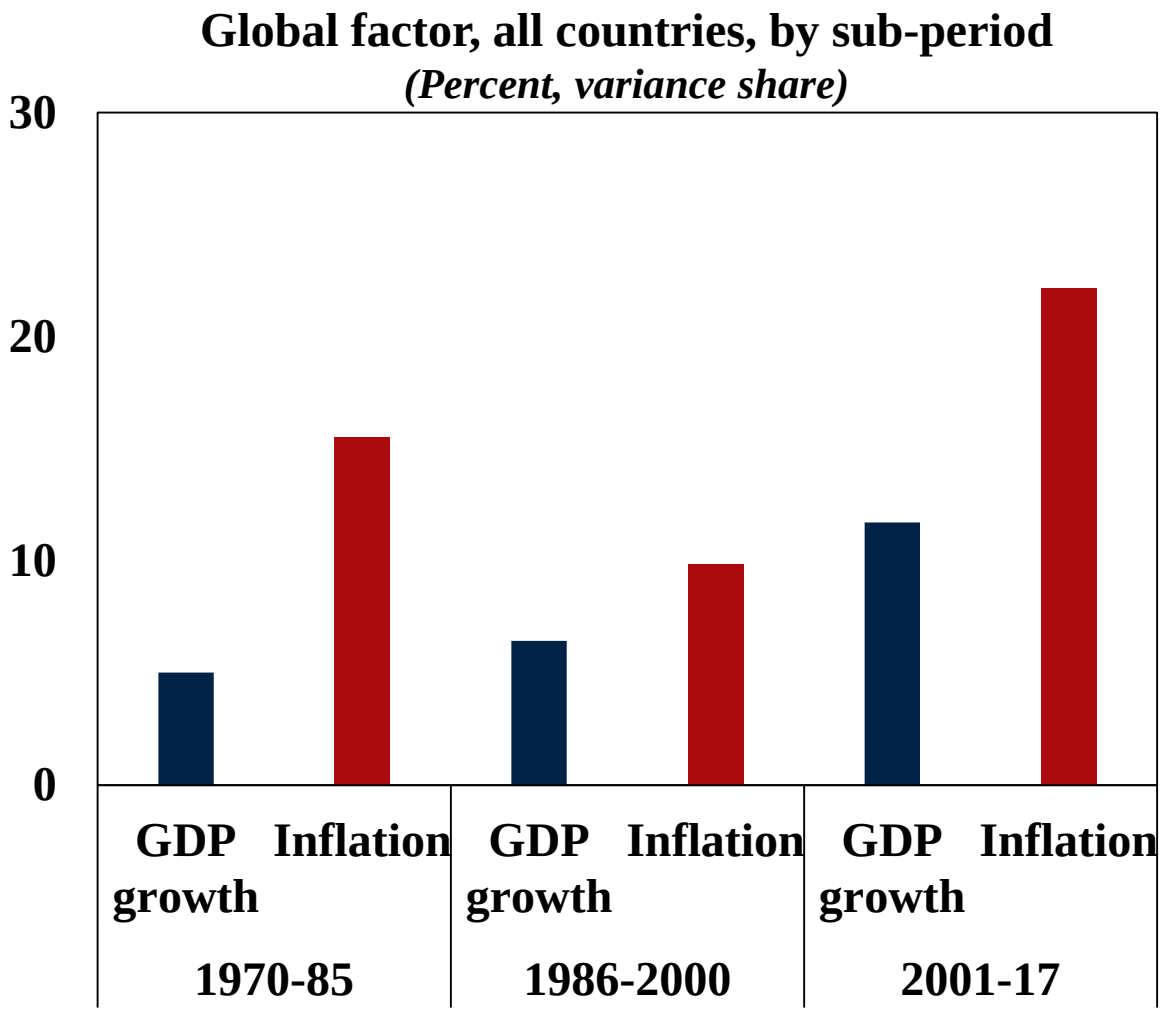
Source: World Bank.
Note: Contributions of global and group factors to inflation variance, estimated with the baseline dynamic factor model (2-factor model with a global factor and a group factor) for the period of 1970-2017 (Left Panel) and for three sub-periods (Right Panel). Median estimates across 99 countries (25 advanced economies and 74 EMDEs including 16 low-income countries).

THE WORLD BANK
IBRD • IDA | WORLD BANK GROUP
Development Economics • Prospects

Inflation versus Output Synchronization

Higher Synchronization in Inflation than Output





Source: World Bank.
Note: Contribution of global factor to variance of real GDP growth and to inflation, estimated with a 2-factor dynamic factor model. Median estimates across 99 economies (25 advanced economies and 74 EMDEs, including 16 low-income countries).

THE WORLD BANK
IBRD • IDA | WORLD BANK GROUP
Development Economics • Prospects

Tradables and Nontradables Inflation

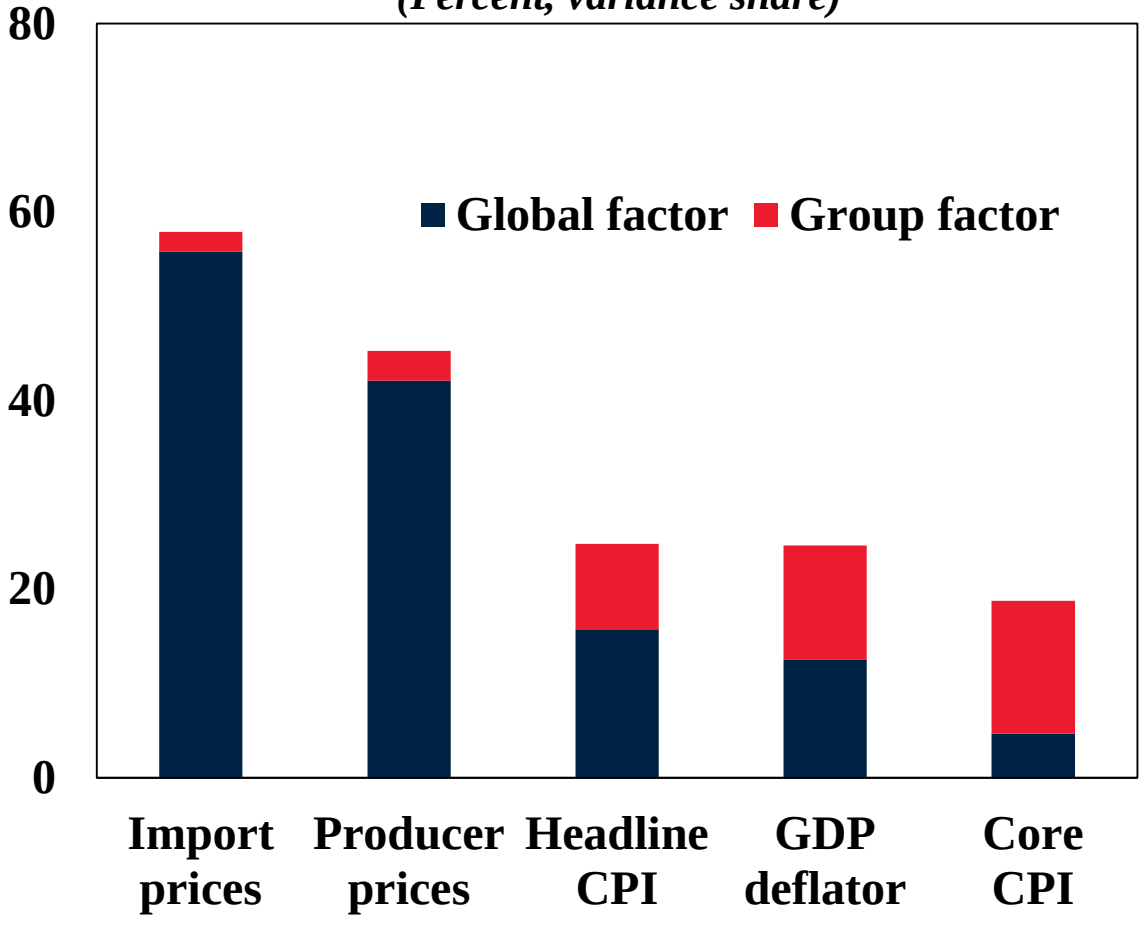
- **Measures of inflation:** Five inflation measures that differ in their tradables content
  - Above-average tradables content: Import prices, producer prices
  - Below-average tradables content: GDP deflator, core CPI
- **Methodology:** Estimate a two-factor dynamic factor model for each measure; and estimate a two-factor model for each group of "tradables" measures (import prices, producer prices, headline consumer prices) and "nontradables" measures (core and headline consumer prices, GDP deflator)
- **Database:** Annual data for 25 advanced economies and 13 EMDEs for 1970-2016

THE WORLD BANK
IBRD • IDA | WORLD BANK GROUP
Development Economics • Prospects

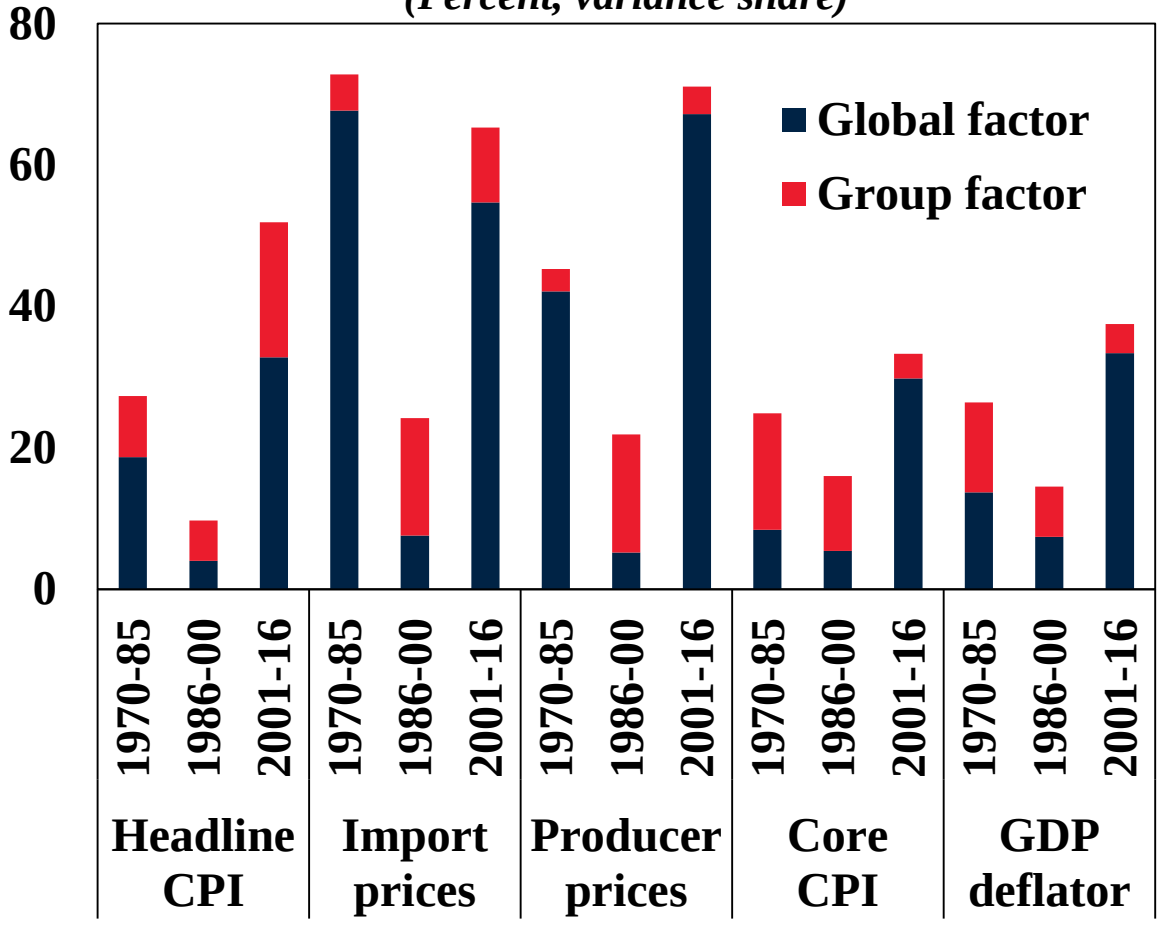
Global Inflation Synchronization by Different Measures

Greater Synchronization with Larger Tradables Content

Global and group factors, 1970-2016
(Percent, variance share)



Global and group factors, by sub-period
(Percent, variance share)

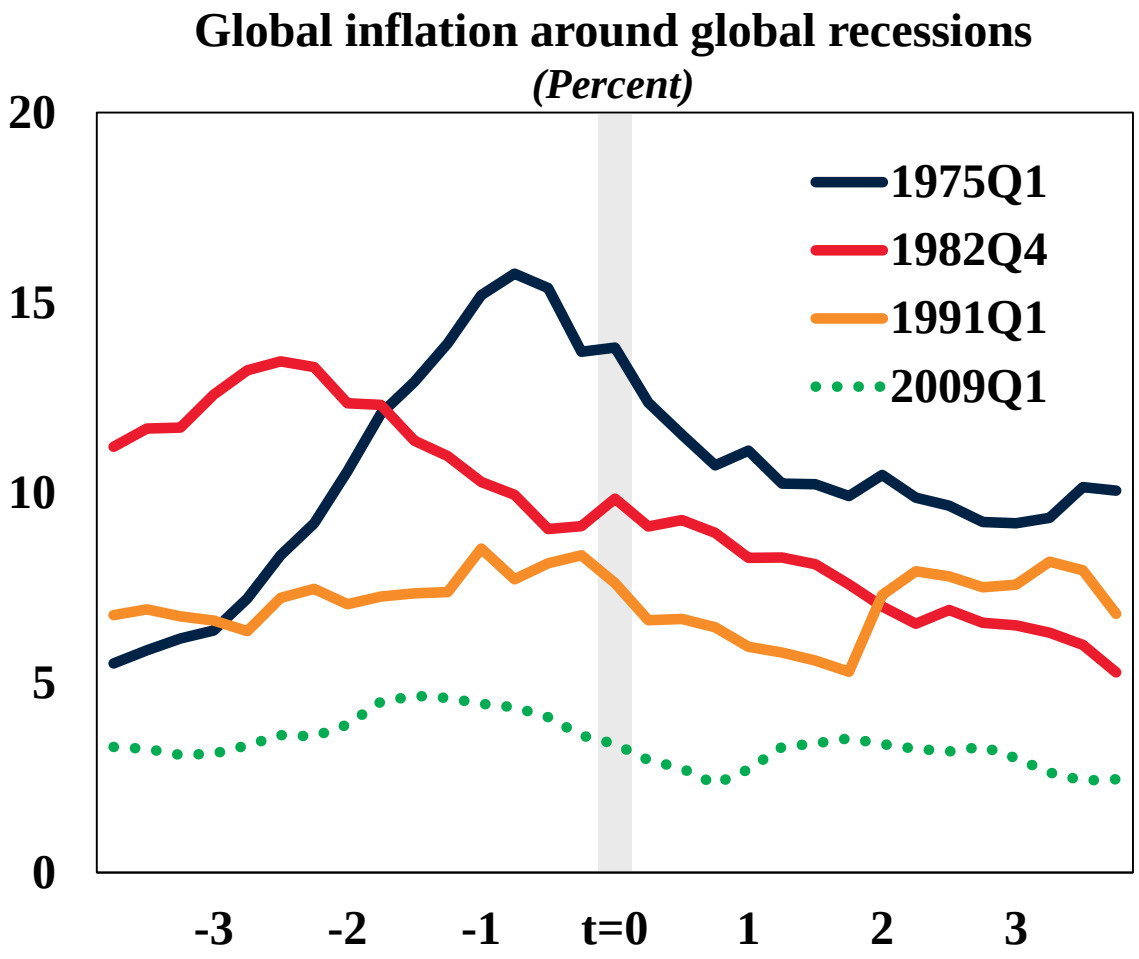


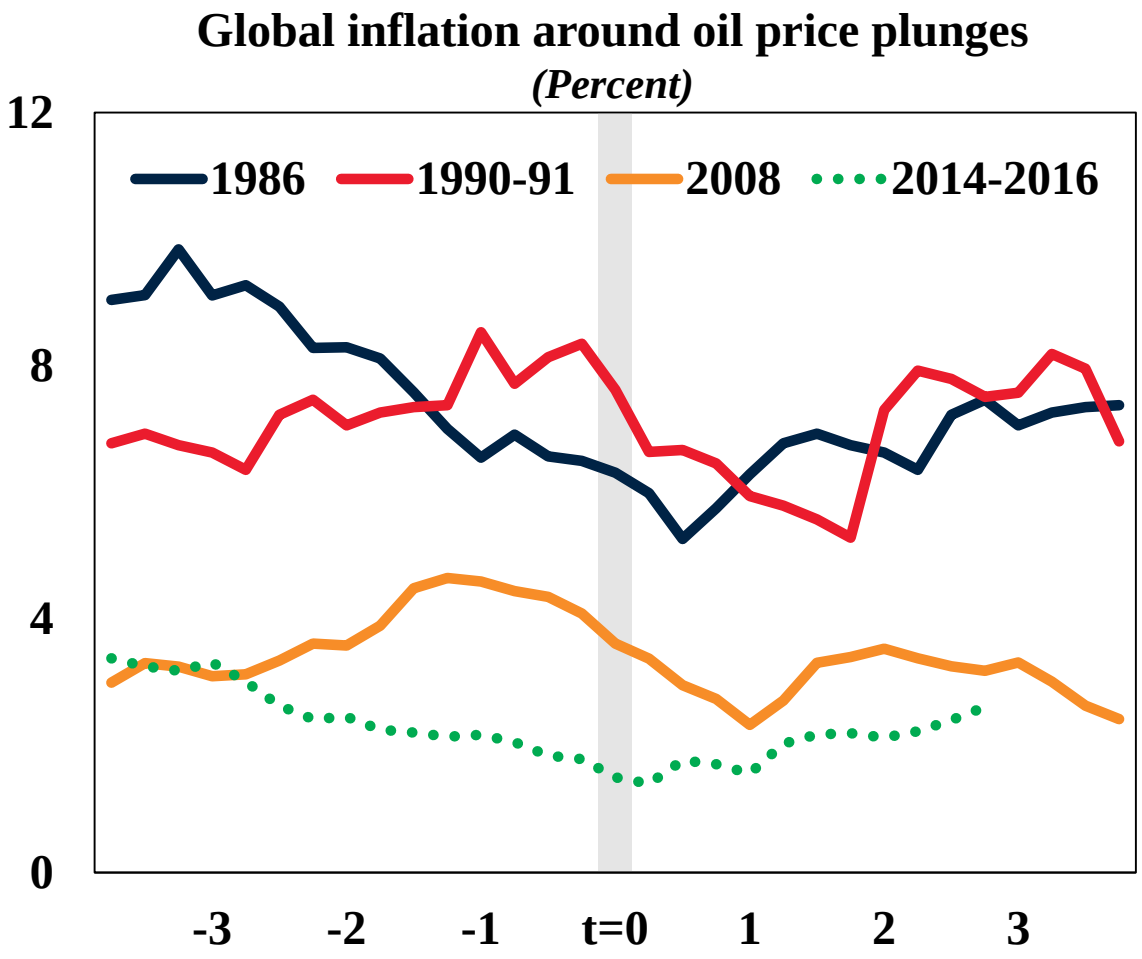
Source: World Bank.
Note: Contributions of global and group factors to variance of various inflation measures, estimated with a baseline dynamic factor model (2-factor model with a global factor and a group factor) using detrended inflation rates in 38 countries (25 advanced economies and 13 EMDEs) for the period of 1970-2016.

THE WORLD BANK
IBRD • IDA | WORLD BANK GROUP
Development Economics • Prospects

Global Inflation and Global Events

Substantial Inflation Movements around Global Events





Sources: Baffes et al. (2015); Haver Analytics; World Bank.

Note: Horizontal axis indicates years before and after the troughs of global recessions (Left Panel) or local troughs of short-term oil price cycle (Right Panel), represented as *t*=0. Global inflation is defined as median trend inflation (9-quarter moving average) across 65 countries, consisting of 25 advanced economies and 40 EMDEs. Left Panel. Troughs of global recessions are identified using global per capita GDP and the algorithm in Harding and Pagan (2002) and are consistent with the results in Kose and Terrones (2015). Right Panel. As Baffes et al. (2015) identify six oil price plunges of more than 30 percent (1986, 1990-91, 1997-98, 2001, 2008-09, and 2014-16), the figure shows the four episodes with the largest oil price plunges. Time *t* refers to: 1986Q1, 1991Q1, 2008Q4, and 2014Q4.

THE WORLD BANK
IBRD • IDA | WORLD BANK GROUP
Development Economics • Prospects

Global and Domestic Shocks Driving Inflation

Factor-Augmented Vector Autoregression (FAVAR)

- **Methodology:** Estimate country-specific Factor-Augmented Structural Vector-Autoregression models (Mumtaz and Surico 2009; Charnavoki and Dolado 2014) to identify the impact of global and domestic shocks on domestic inflation

$$B_0 y_t = \alpha + \sum_{i=1} B_i y_{t-i} + \varepsilon_t$$

ε_t : vector of orthogonal structural innovations

y_t : global inflation ($f_t^{\pi, global}$), global output growth ($f_t^{Y, global}$), oil price growth (Δop), domestic inflation (π^i_t), output growth (Y^i_t), changes in NEER (XR^i_t), short-term interest rates (I^i_t)

Global inflation and output growth proxied by global factors from the dynamic factor model

$$Y^i = \beta_{global} f_t^{Y,global} + e^{Y,i}$$

$$\pi^i = \beta_{global} f_t^{\pi,global} + e^{\pi,i}$$

- **Database:** 29 advanced economies and 26 EMDEs for 1970Q1-2017Q4

THE WORLD BANK
IBRD • IDA | WORLD BANK GROUP
Development Economics • Prospects

Identification of Global and Domestic Shocks - 1

Sign Restrictions

- Identification using sign restrictions (Charnavoki and Dolado 2014; and Forbes, Hjortsoe, and Nenova 2017, 2018)
- Two-step identification: Global and domestic blocks

Identification of global and domestic shocks

$$\begin{bmatrix} u_t^{Y,global} \\ u_t^{OP} \\ u_t^{\pi,global} \\ u_t^{Y,domestic} \\ u_t^{\pi,domestic} \\ u_t^{I,domestic} \\ u_t^{ER} \end{bmatrix} = \begin{bmatrix} + & - & + & 0 & 0 & 0 & 0 \\ + & + & + & 0 & 0 & 0 & 0 \\ + & + & - & 0 & 0 & 0 & 0 \\ * & * & * & + & + & - & * \\ * & * & * & + & - & - & * \\ * & * & * & * & * & + & * \\ * & * & * & * & * & + & + \end{bmatrix} \begin{bmatrix} \varepsilon_t^{\text{Global demand}} \\ \varepsilon_t^{\text{Oil price}} \\ \varepsilon_t^{\text{Global supply}} \\ \varepsilon_t^{\text{Domestic demand}} \\ \varepsilon_t^{\text{Domestic supply}} \\ \varepsilon_t^{\text{Monetary policy}} \\ \varepsilon_t^{\text{Exchange rate}} \end{bmatrix}$$

THE WORLD BANK
IBRD • IDA | WORLD BANK GROUP
Development Economics • Prospects

Identification of Global and Domestic Shocks - 2

1

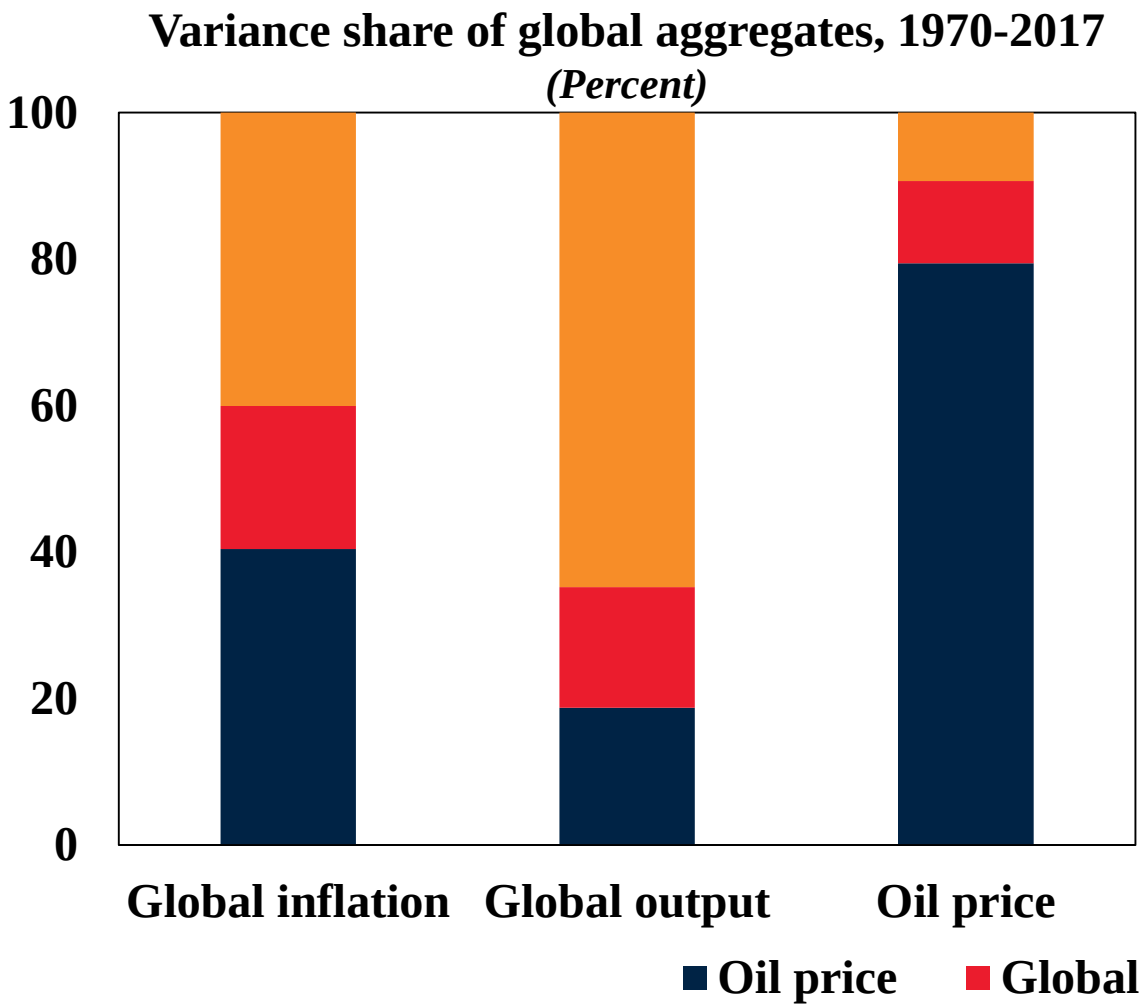
Global shocks

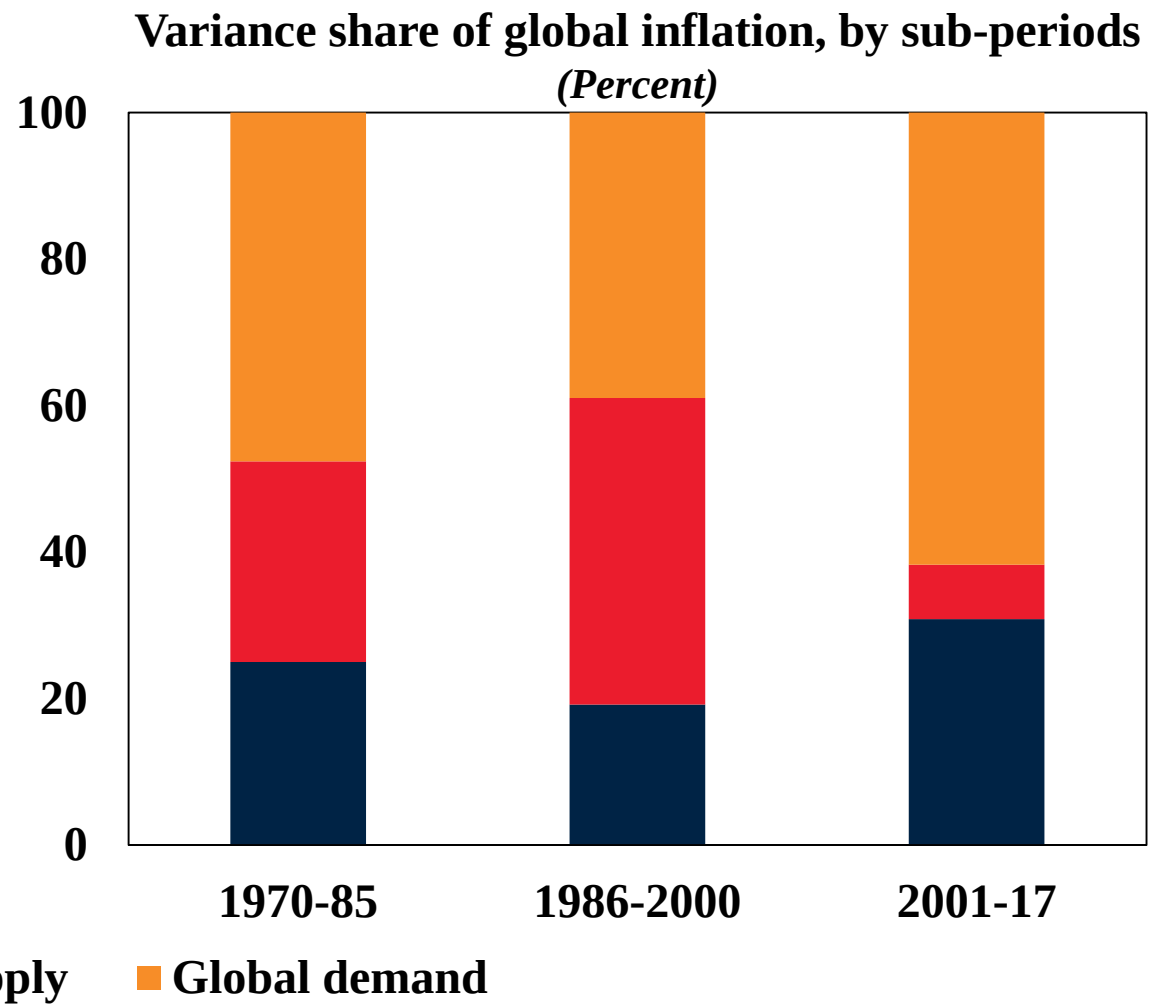
- *Global demand shocks raise global inflation, global output growth and oil prices.*
- *Global supply shocks reduce global inflation but raise global output growth and oil prices.*
- *Oil price shocks raise oil prices and global inflation but reduce global output growth.*

THE WORLD BANK
IBRD • IDA | WORLD BANK GROUP
Development Economics • Prospects

Drivers of Global Inflation

Driven by Global Demand and Oil Price Shocks; Demand Shocks Becoming More Important





Oil price | Global supply | Global demand

Source: World Bank.
Note: Variance decompositions estimated by a global factor-augmented vector autoregressive (FAVAR) model, based on 29 advanced economies and 26 EMDEs over 1970-2017 (Left Panel) and during its sub-periods (Right Panel).

THE WORLD BANK
IBRD • IDA | WORLD BANK GROUP
Development Economics • Prospects

Identification of Global and Domestic Shocks - 3

1 Global shocks

- *Global demand shocks raise global inflation, global output growth and oil prices.*
- *Global supply shocks reduce global inflation but raise global output growth and oil prices.*
- *Oil price shocks raise oil prices and global inflation but reduce global output growth.*

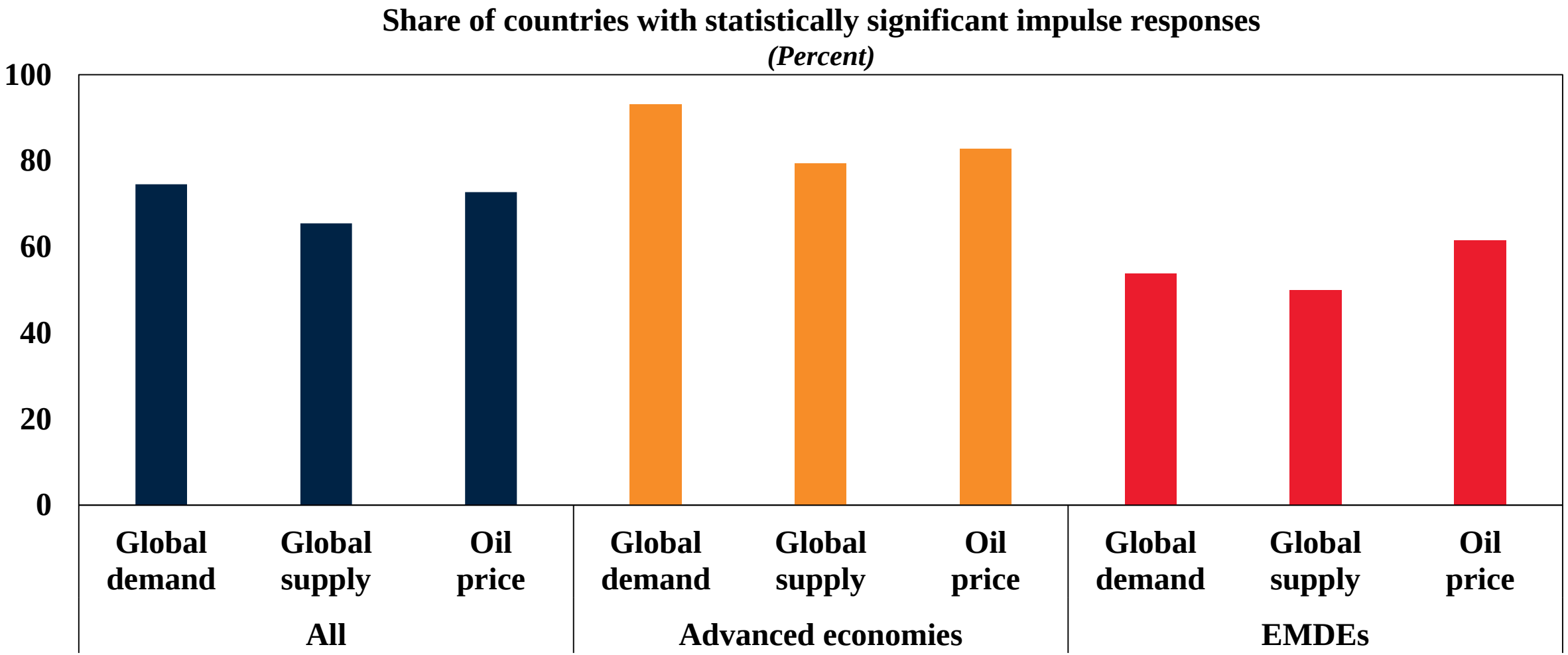
2 Domestic shocks

- *Global shocks affect domestic variables contemporaneously, but vice versa only with a one-quarter delay.*
- *Domestic demand shocks raise inflation and output growth.*
- *Domestic supply shocks reduce inflation but raise output growth.*
- *Contractionary monetary policy shocks raise interest rates, reduce inflation and output growth, and appreciate exchange rates.*

THE WORLD BANK
IBRD • IDA | WORLD BANK GROUP
Development Economics • Prospects

Impact of Global Shocks on Domestic Inflation

Significant in Most Countries

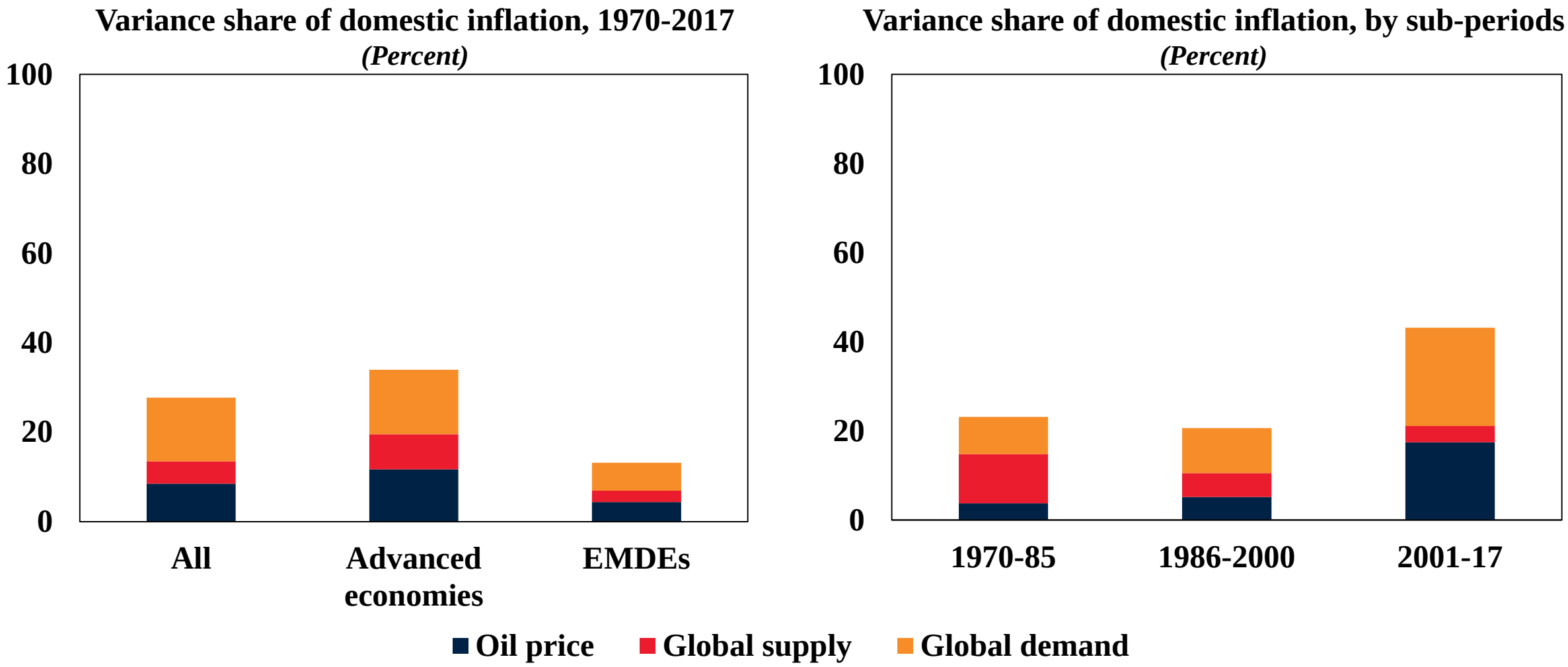


Source: World Bank.
Note: Share of countries with statistically significant cumulative impulse responses of domestic inflation after two years to a one-standard-deviation shock to global demand, global supply, and oil prices. Estimated results are considered to be statistically significant if they are within 16-84 percent confidence band.

THE WORLD BANK
IBRD • IDA | WORLD BANK GROUP
Development Economics • Prospects

Contributions of Global Shocks to Domestic Inflation

One-Quarter of Domestic Inflation Variance, and Growing

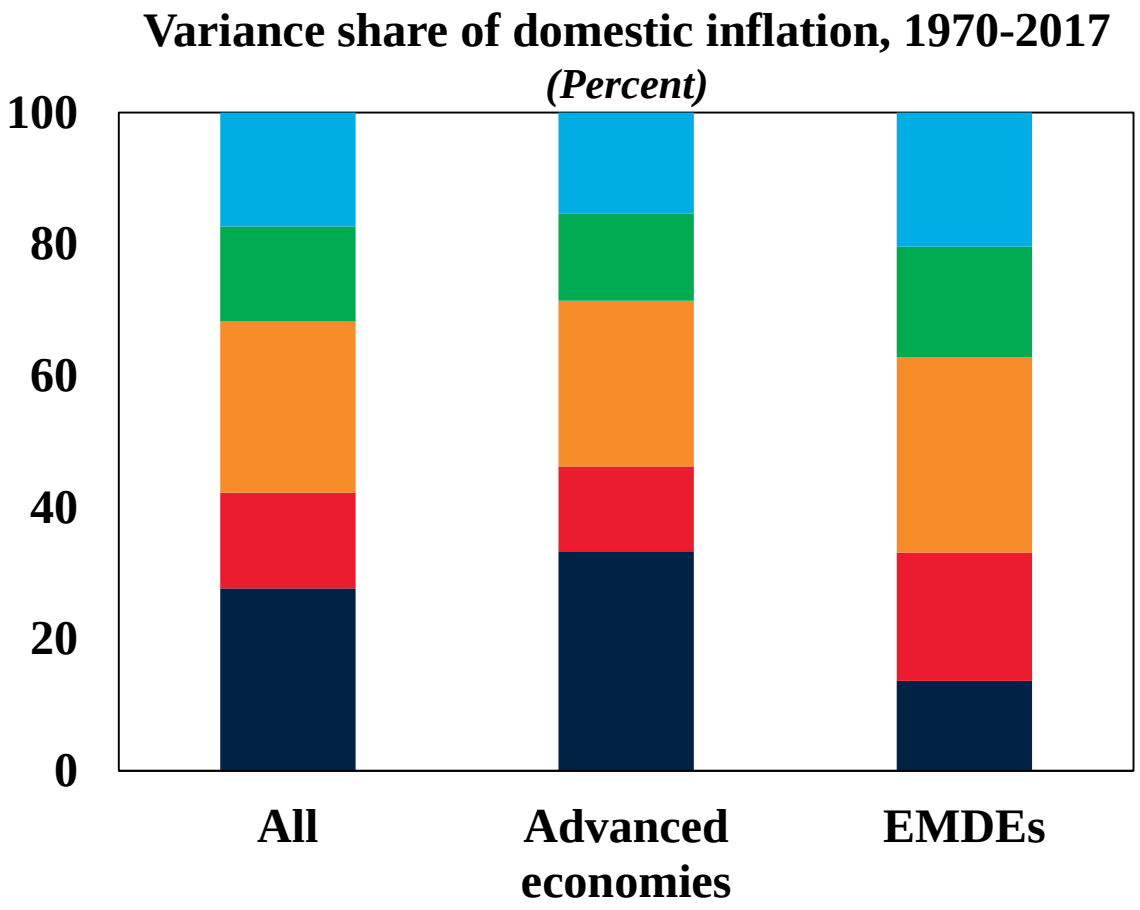


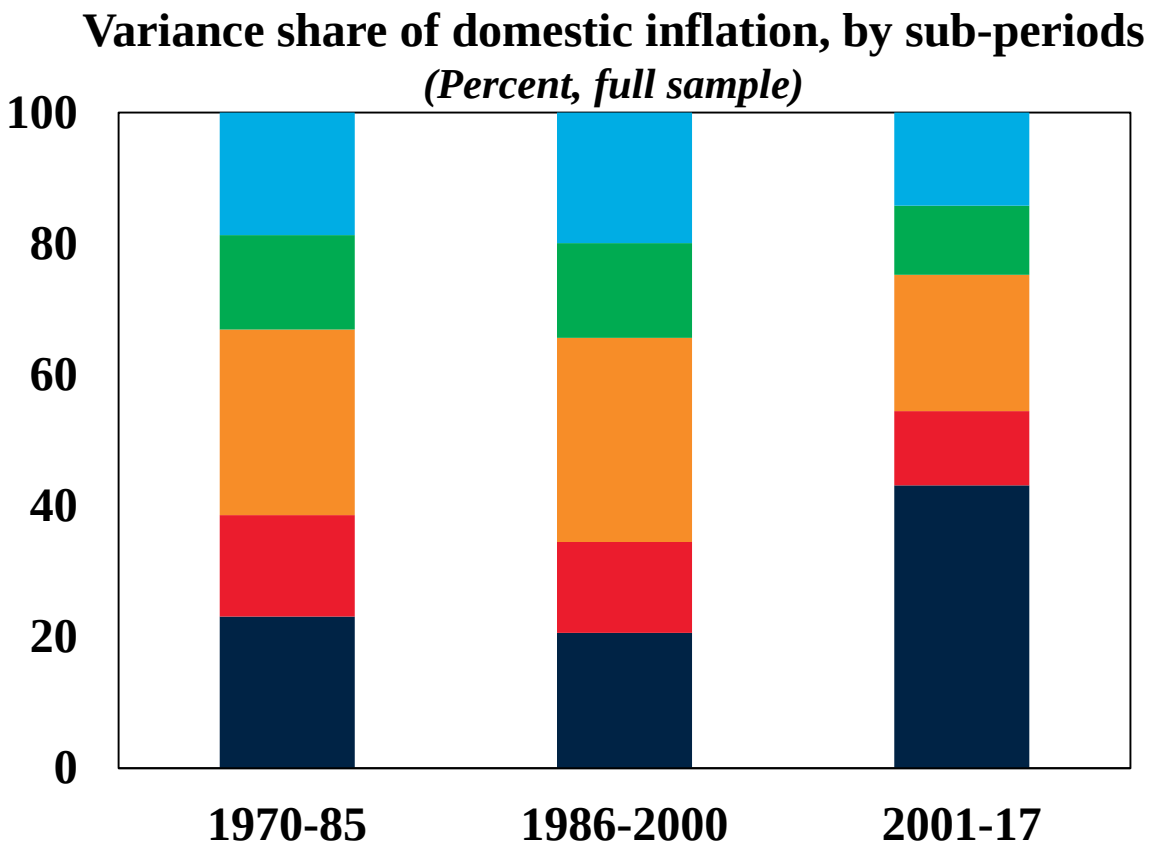
Source: World Bank.
Note: Median shares of country-specific inflation variance accounted for by global shocks (i.e., global demand, global supply, and oil prices) based on country-specific factor-augmented vector autoregressive (FAVAR) models for 29 advanced economies and 26 EMDEs for 1970-2017 (Left Panel) and for three sub-sample periods (Right Panel).

THE WORLD BANK
IBRD • IDA | WORLD BANK GROUP
Development Economics • Prospects

Contributions of Domestic Shocks to Domestic Inflation

Domestic Shocks Contribute Substantially but Becoming Less Important





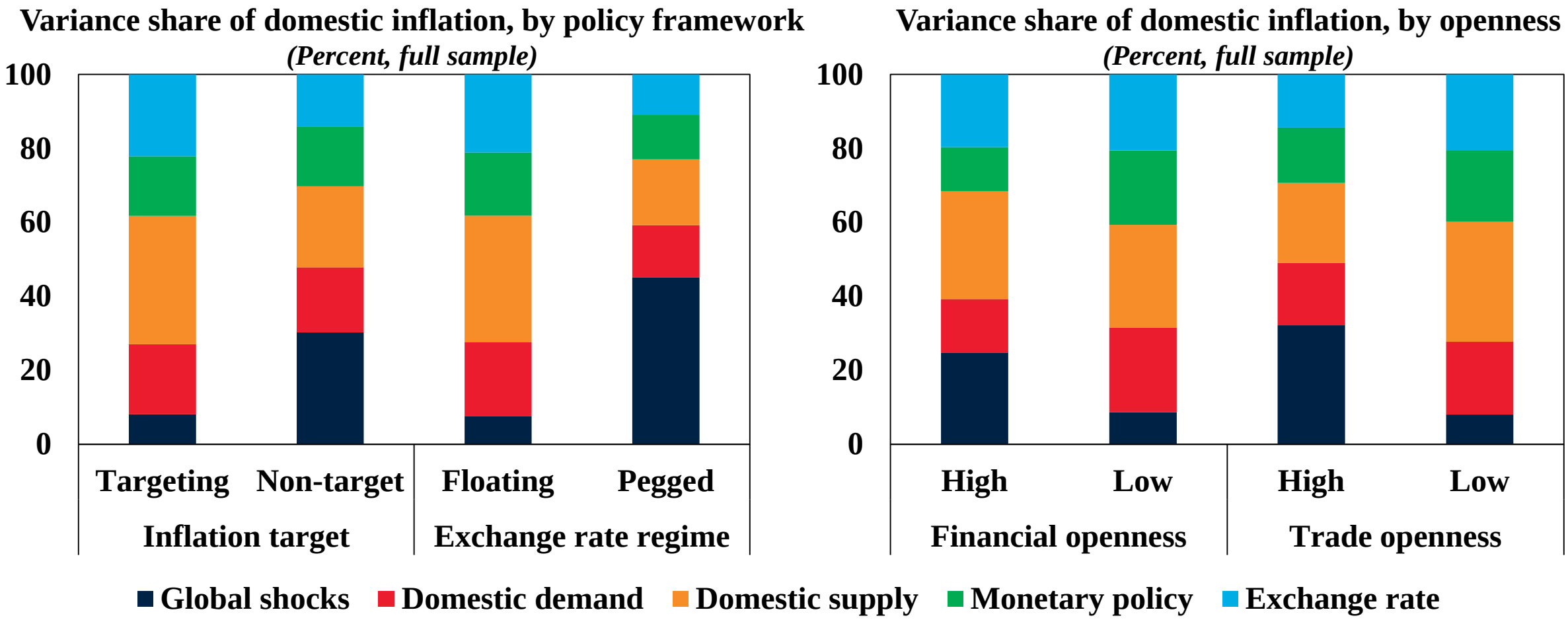
■ Global shocks ■ Domestic demand ■ Domestic supply ■ Monetary policy ■ Exchange rate

Source: World Bank.
Note: Median shares of country-specific inflation variance accounted for by global shocks and domestic shocks (i.e., domestic demand, domestic supply, monetary policy (interest rates), and exchange rates) based on country-specific factor-augmented vector autoregressive (FAVAR) models. The models are estimated for 29 advanced economies and 26 EMDEs for 1970-2017 (Left Panel) and for three sub-sample periods (Right Panel).

THE WORLD BANK
IBRD • IDA | WORLD BANK GROUP
Development Economics • Prospects

Contributions of Global and Domestic Shocks

Domestic Shocks More Important in Less Open Economies



Sources: Chinn and Ito (2006); Ilzetzki, Reinhart, and Rogoff (2017); IMF (2016); World Bank.

Note: Median shares of country-specific inflation variance accounted for by global shocks and domestic shocks (i.e., domestic demand, domestic supply, monetary policy (interest rates), and exchange rates) based on country-specific factor-augmented vector autoregressive (FAVAR) models, estimated for 29 advanced economies and 26 EMDEs for 1970-2017. Left Panel. Inflation targeting regimes are defined as in IMF (2016). Flexible exchange rate regimes ("Floating") are defined as freely floating and managed floating exchange rate regimes (Ilzetzki, Reinhart, and Rogoff 2017), and all other regimes are defined as pegged exchange rate regimes. Right Panel. Countries with "high" and "low" financial openness are, respectively, those above and below median of the capital account openness index in Chinn and Ito (2006). Countries with "high" and "low" trade openness are defined as those with trade-to-GDP ratios above and below median, respectively.

THE WORLD BANK
IBRD • IDA | WORLD BANK GROUP
Development Economics • Prospects

Four Questions

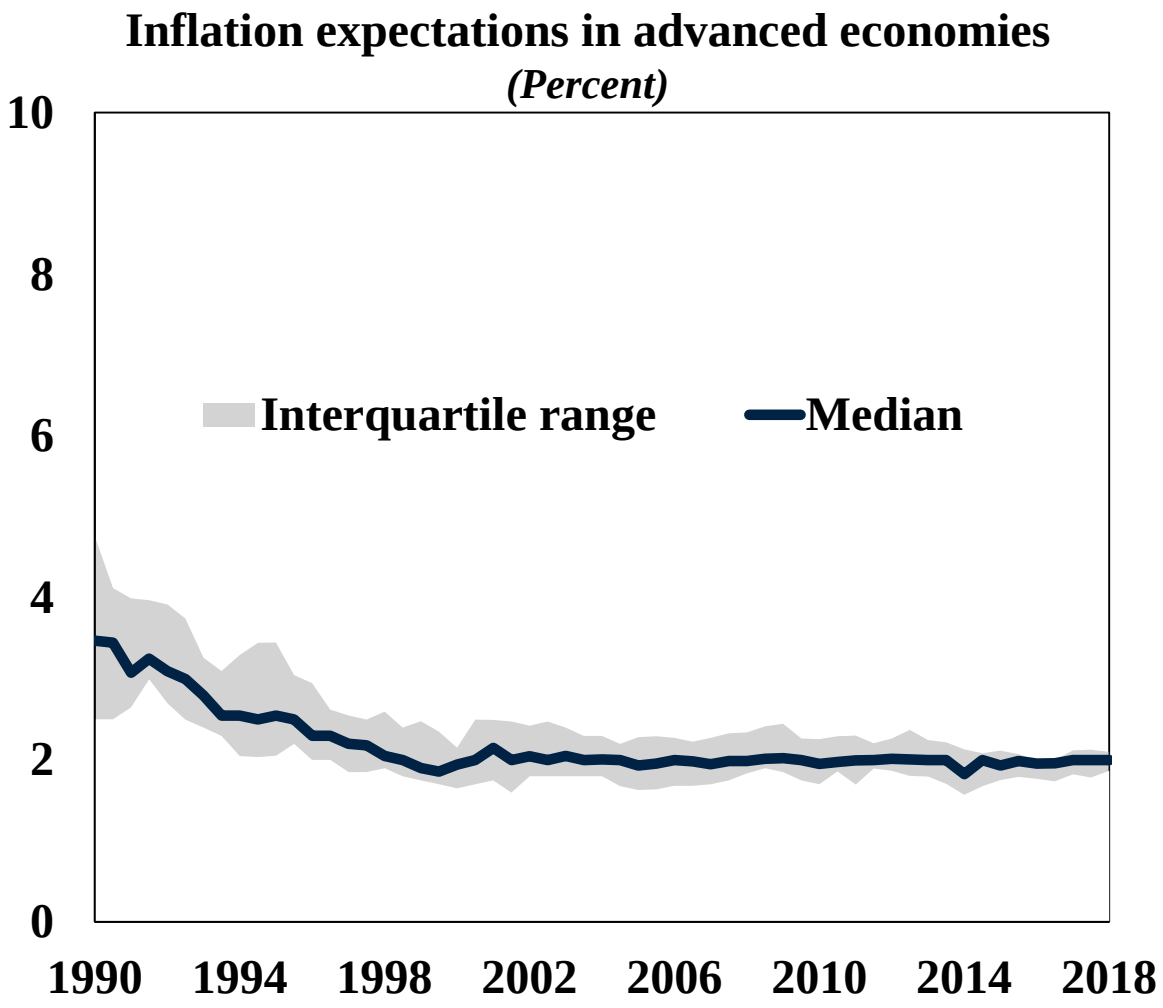
1 **How has inflation evolved in EMDEs?** *Declined over time thanks to the sharp fall in global inflation, supported by cyclical and structural developments. If momentum for these developments wanes, maintaining low inflation may become as difficult as achieving it.*

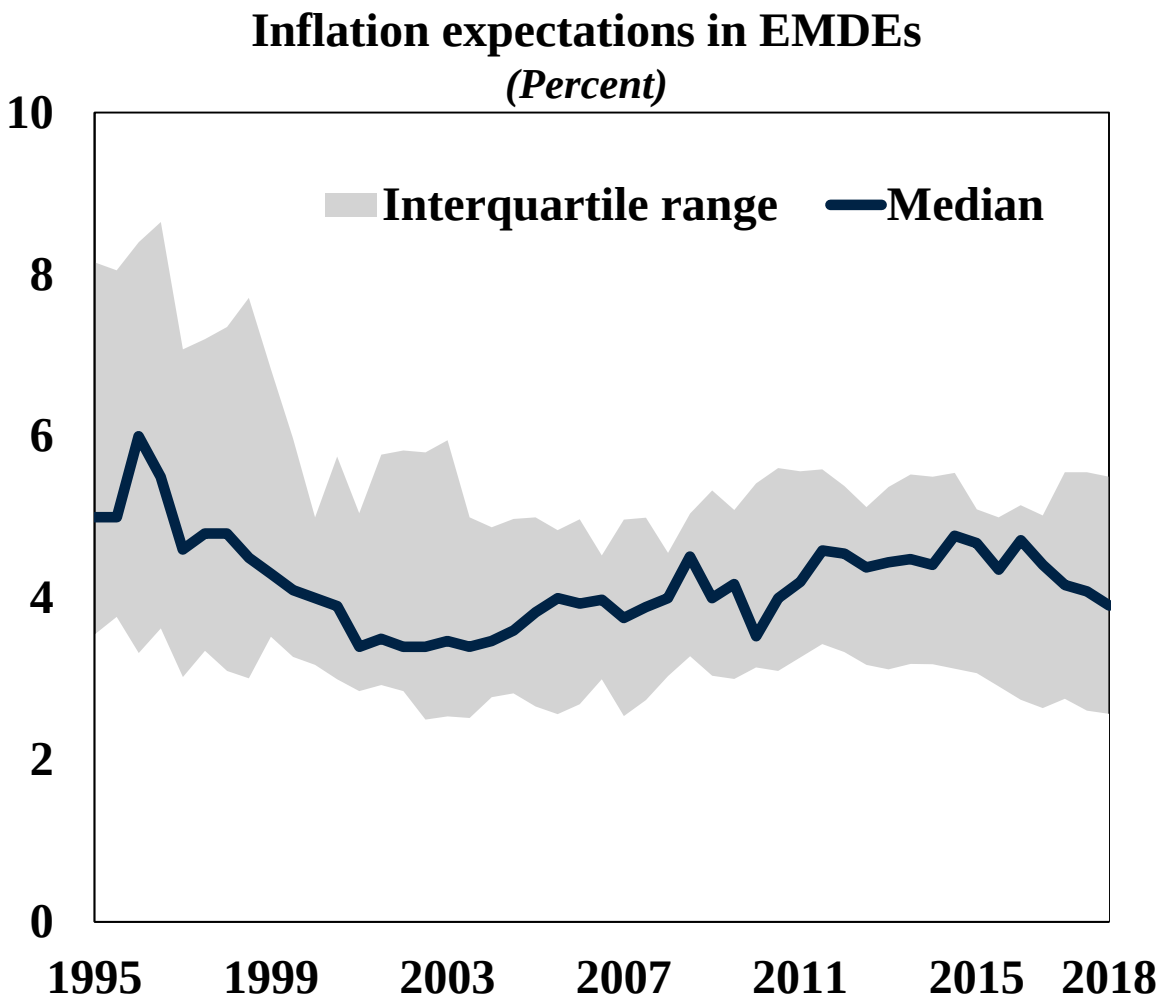
2 **What have been the global and domestic drivers of inflation?** *A wide range of shocks… A global inflation cycle has emerged and inflation synchronization has strengthened among EMDEs over time. Global shocks have become more important over time but domestic shocks remain the main source of national inflation variation.*

3 **How well-anchored are EMDE inflation expectations?** *Despite becoming better-anchored since the 1990s, inflation expectations remain less well anchored in EMDEs than advanced economies. Stronger monetary frameworks and central bank independence are associated with better anchoring of expectations and lower exchange rate pass-through.*

Inflation Expectations

Broad-Based Decline, but Remain Higher in EMDEs





Sources: Consensus Economics, International Monetary Fund, World Bank.
Note: Inflation expectations refer to 5-year-ahead expectations of annual inflation and measured at a bi-annual frequency. Interquartile range of the country sample. Sample includes 24 advanced economies (over 1990H1-2018H1) and 23 EMDEs (over 1995H1-2018H1).

THE WORLD BANK
IBRD • IDA | WORLD BANK GROUP
Development Economics • Prospects

Anchoring Inflation Expectations

Methodology and Database

- Methodology

  Degree of anchoring. Regression of changes in five-year-ahead inflation expectations on inflation shocks (fixed-effects panel regression and country-level time-varying models). Inflation shock: difference between inflation and short-term inflation expectations six months prior

  Roles of inflation shocks. *Same regression above but incorporating*

  - Global inflation shock: First principal component of inflation shocks for the full sample
  - Domestic inflation shock: Residual from a regression of inflation shock on global inflation shock

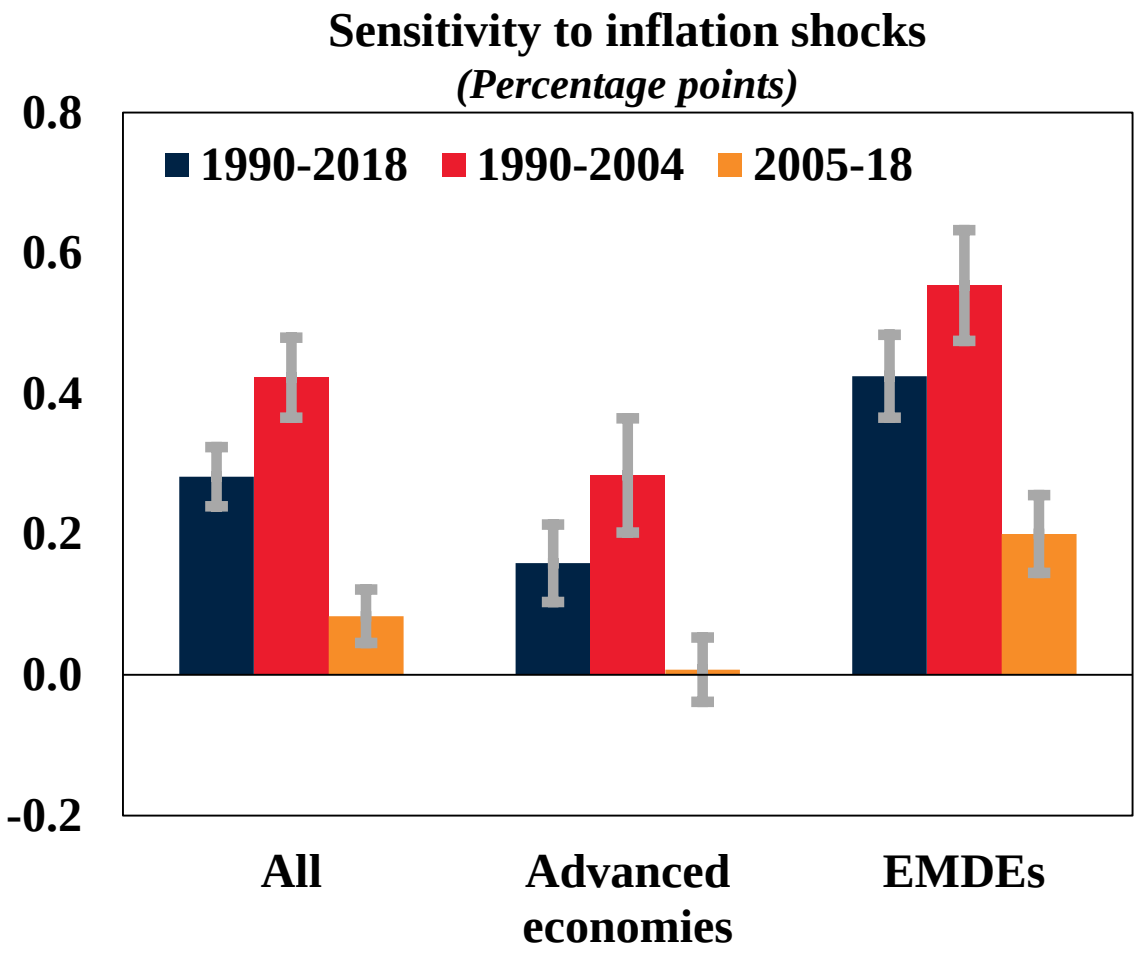
  Determinants of the degree of anchoring. Regression of the estimated sensitivity by a time-varying model for each country on different determinants (monetary frameworks, central bank independence, exchange rate regimes, globalization, and fiscal variables)

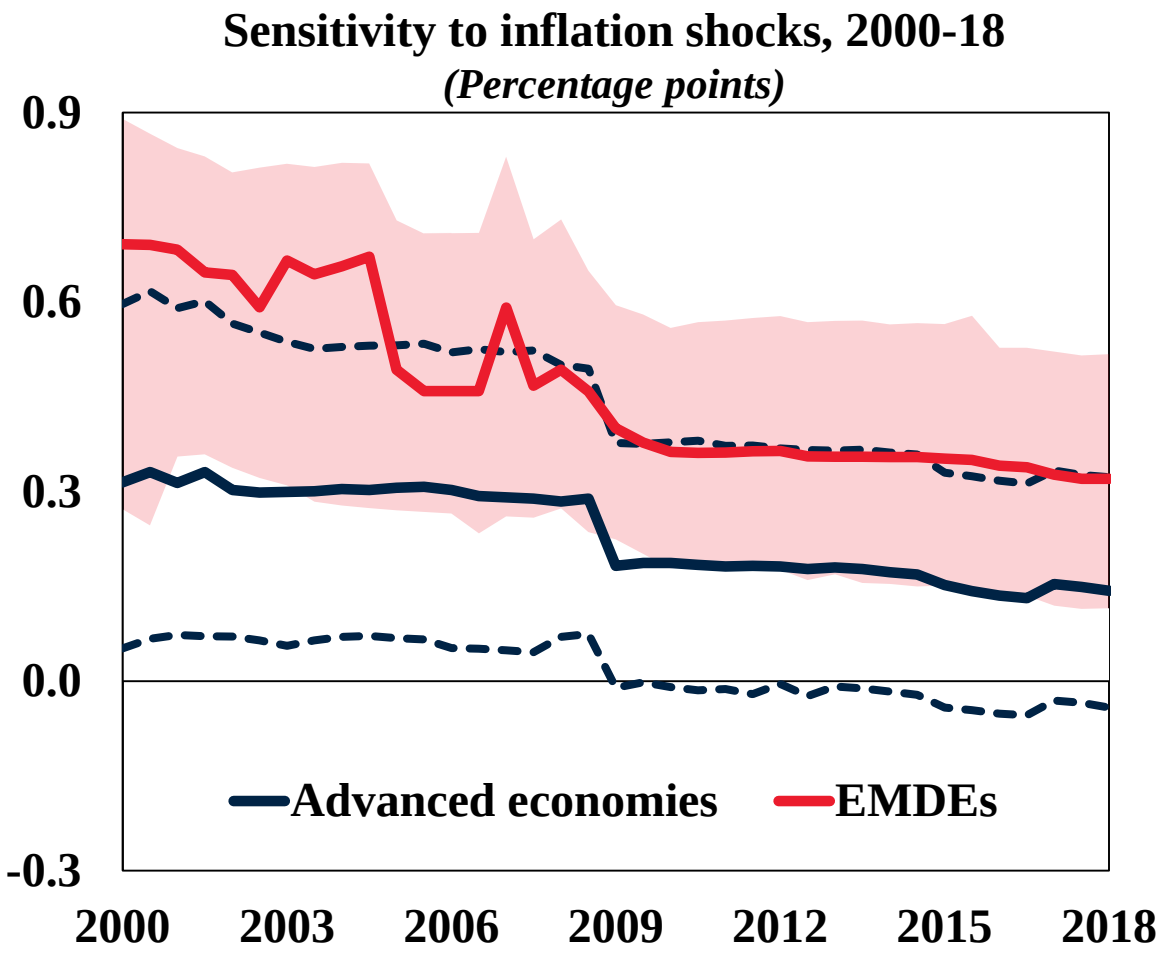
- Database: 24 advanced economies and 23 EMDEs
  1990H2-2018H1: Analysis of inflation shocks
  1995-2016: Analysis of determinants of anchoring

THE WORLD BANK
IBRD • IDA | WORLD BANK GROUP
Development Economics • Prospects

Anchoring Inflation Expectations in EMDEs - 1

Better Anchored Now; Still Not as Well as in Advanced Economies





Sources: Consensus Economics, International Monetary Fund, World Bank.

Note: Inflation shocks are defined as he difference between realized inflation and short-term inflation expectations in the previous period. Left Panel. Sensitivity is estimated using a panel regression of changes in 5-year-ahead inflation expectations on inflation shocks. Bars denote median estimates and vertical lines denote 90 percent confidence intervals. Based on 24 advanced economies and 23 EMDEs for 1990H2-2018H1. Right Panel. Time-varying sensitivity is estimated by regressing changes in 5-year-ahead inflation expectations on inflation shocks. Solid lines denote median estimates and areas between two dotted blue lines and shaded in pink indicate, respectively, medians of 68 percent confidence intervals for advanced economies and EMDEs. Sample includes 24 advanced economies (over 1995H1-2018H1) and 23 EMDEs (over 2000H1-2018H1).

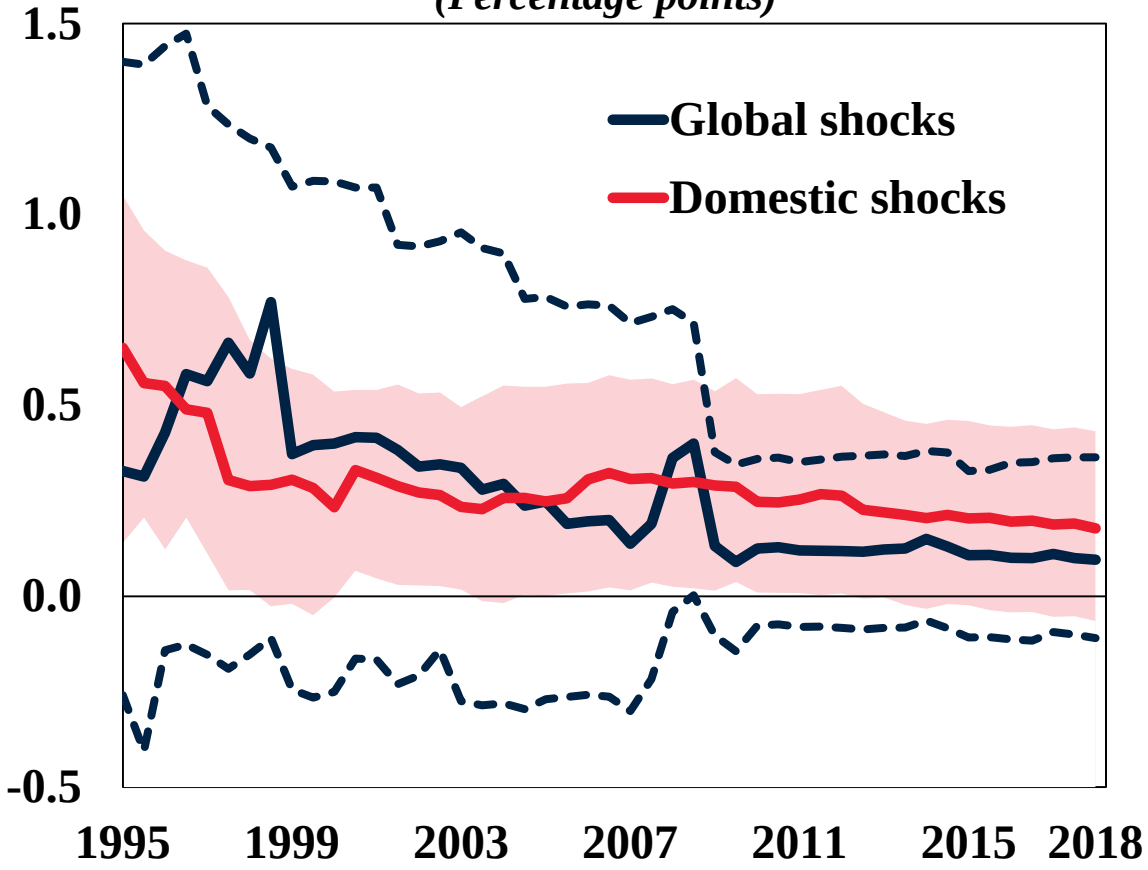
THE WORLD BANK
IBRD • IDA | WORLD BANK GROUP
Development Economics • Prospects

Anchoring Inflation Expectations in EMDEs- 2

Poorly Anchored for Both Domestic and Global Shocks

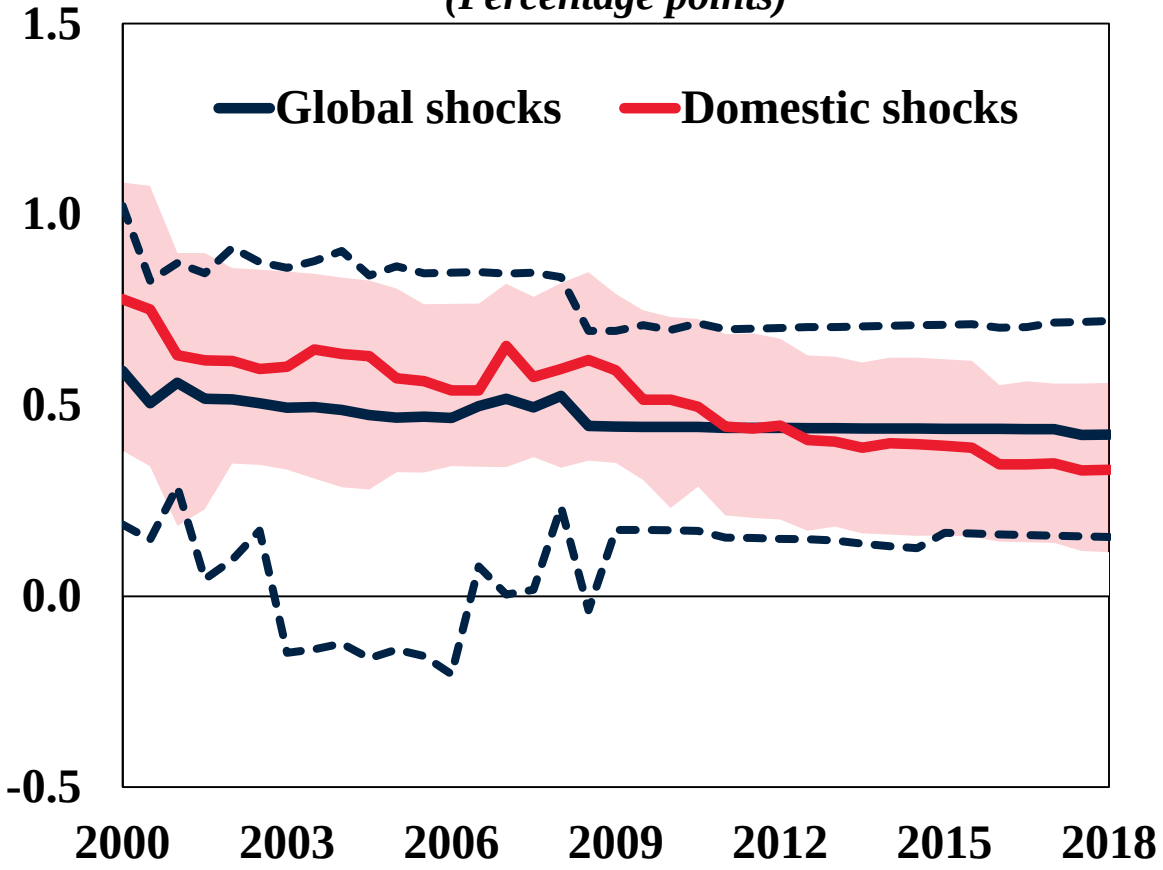
Sensitivity to inflation shocks in advanced economies

(Percentage points)



Sensitivity to inflation shocks in EMDEs

(Percentage points)



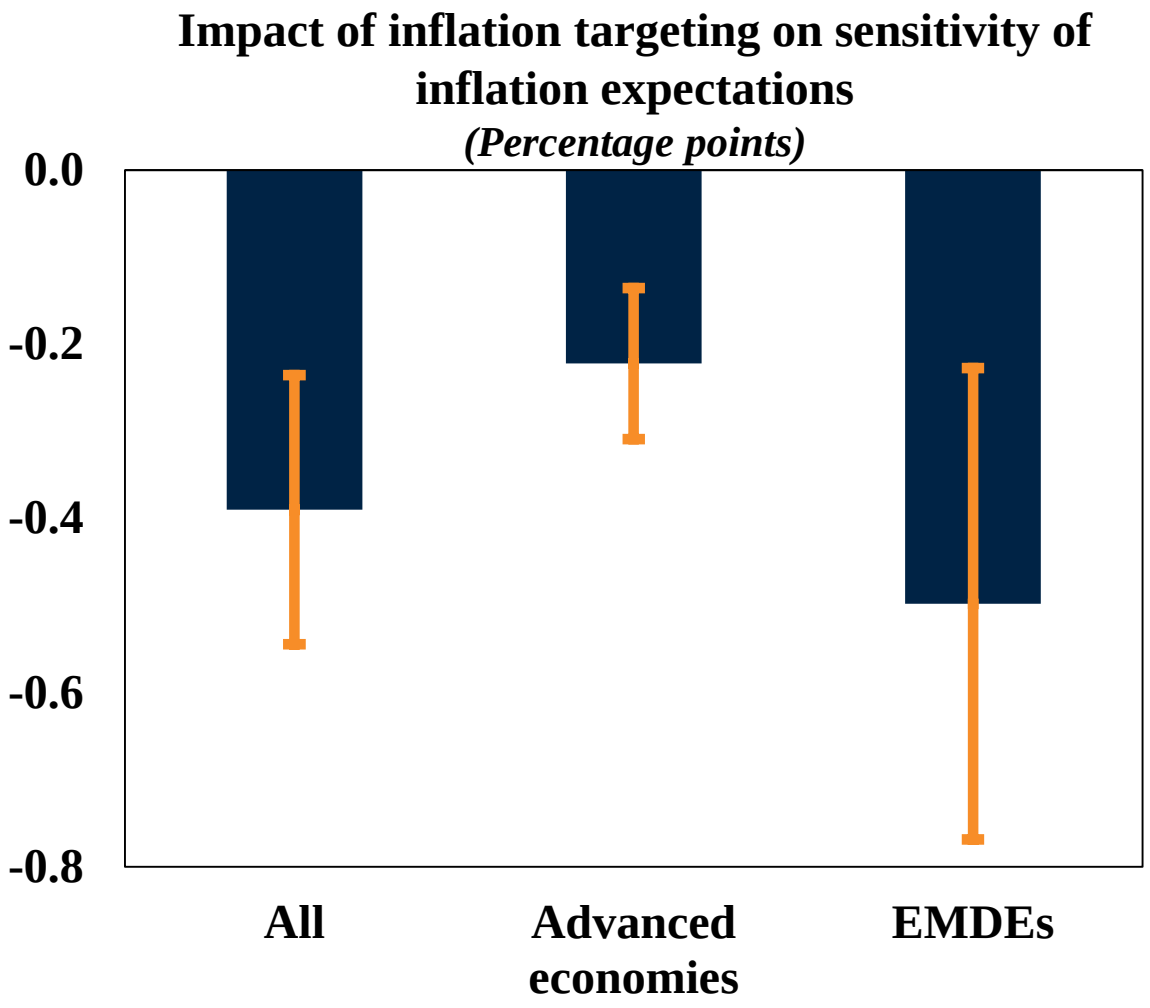
Sources: Consensus Economics, International Monetary Fund, World Bank.

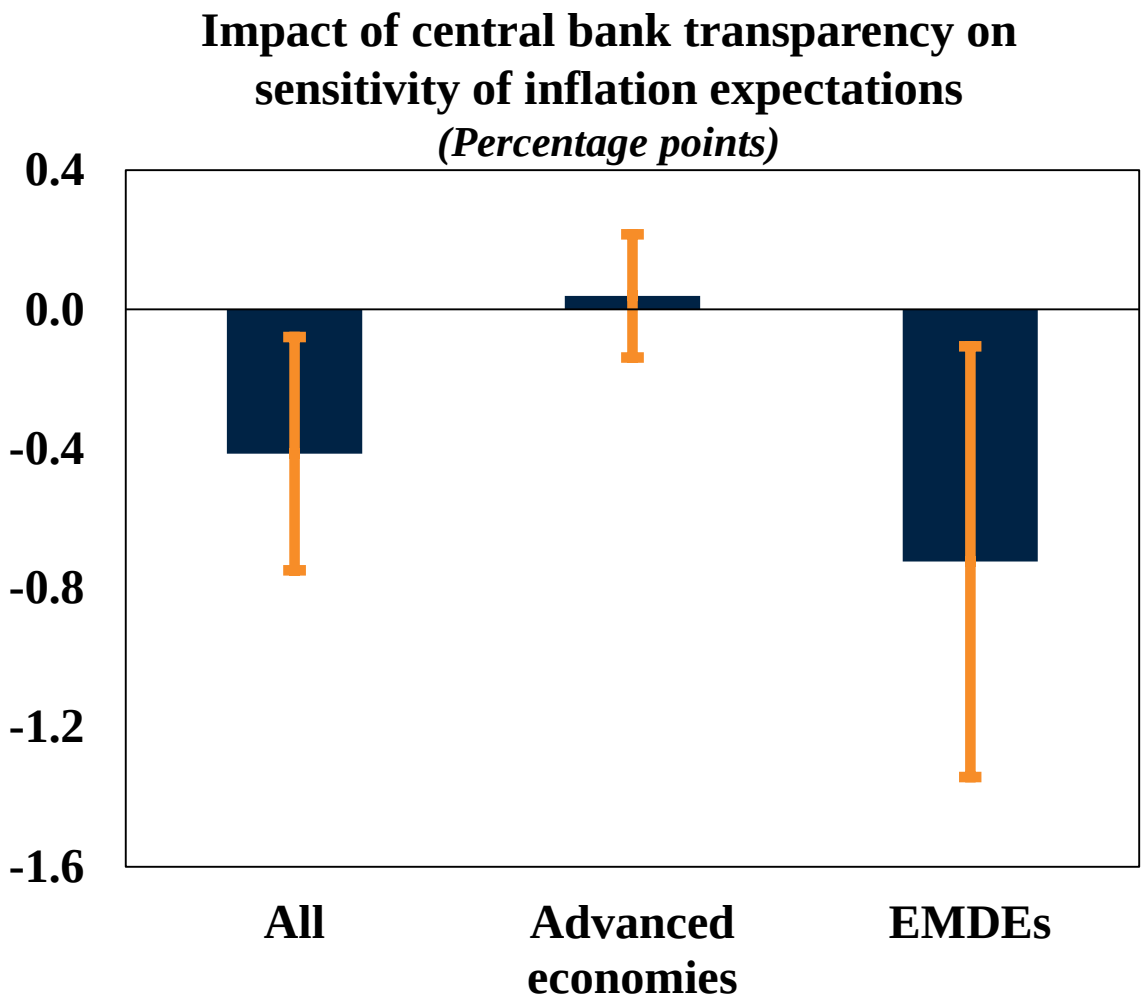
Note: The global inflation shocks are defined as the first principal component of inflation shocks for the full sample of 24 advanced economies over 1990H2-2018H1 and 23 EMDEs over 1995H1-2018H1. The domestic inflation shocks are defined as the residual from a regression of the inflation shocks on the global inflation shocks. Time-varying sensitivity is estimated by regressing changes in 5-year-ahead inflation expectations on global and domestic inflation shocks. Solid lines denote median estimates and areas between two dotted blue lines and shaded in pink indicate, respectively, medians of 68 percent confidence intervals for global shocks and domestic shocks. Sample includes 24 advanced economies (over 1995H1-2018H1) and 23 EMDEs (over 2000H1-2018H1).

THE WORLD BANK
IBRD • IDA | WORLD BANK GROUP
Development Economics • Prospects

Determinants of Anchoring Inflation Expectations

Lower Sensitivity with Inflation-Targeting Regime and Higher Central Bank Transparency





Sources: Consensus Economics; Dincer and Eichengreen (2014); International Monetary Fund, World Bank.

Note: Bars represent coefficients in panel regressions of 47 countries (including 24 advanced economies and 23 EMDEs), based on annual data over 1995-2016. Vertical lines denote 90 percent confidence intervals. Left Panel. Inflation targeting is a dummy variable taking a value of one in countries with an inflation-targeting framework. Right Panel. Central bank transparency is measured by the index of central bank independence and transparency in Dincer and Eichengreen (2014), which ranges from 0 (least independent and transparent) to 15 (most independent and transparent).

THE WORLD BANK
IBRD • IDA | WORLD BANK GROUP
Development Economics • Prospects

Exchange Rate Pass-Through - 1

Factor-Augmented Vector Autoregression (FAVAR)

- **Exchange rate pass-through ratio:** Defined as percent increase in consumer prices associated with a 1 percent nominal effective exchange rate depreciation after one year

$$\text{Pass-Through Ratio} = -\frac{\text{Impulse response of national inflation}}{\text{Impulse response of nominal effective exchange rate}}$$

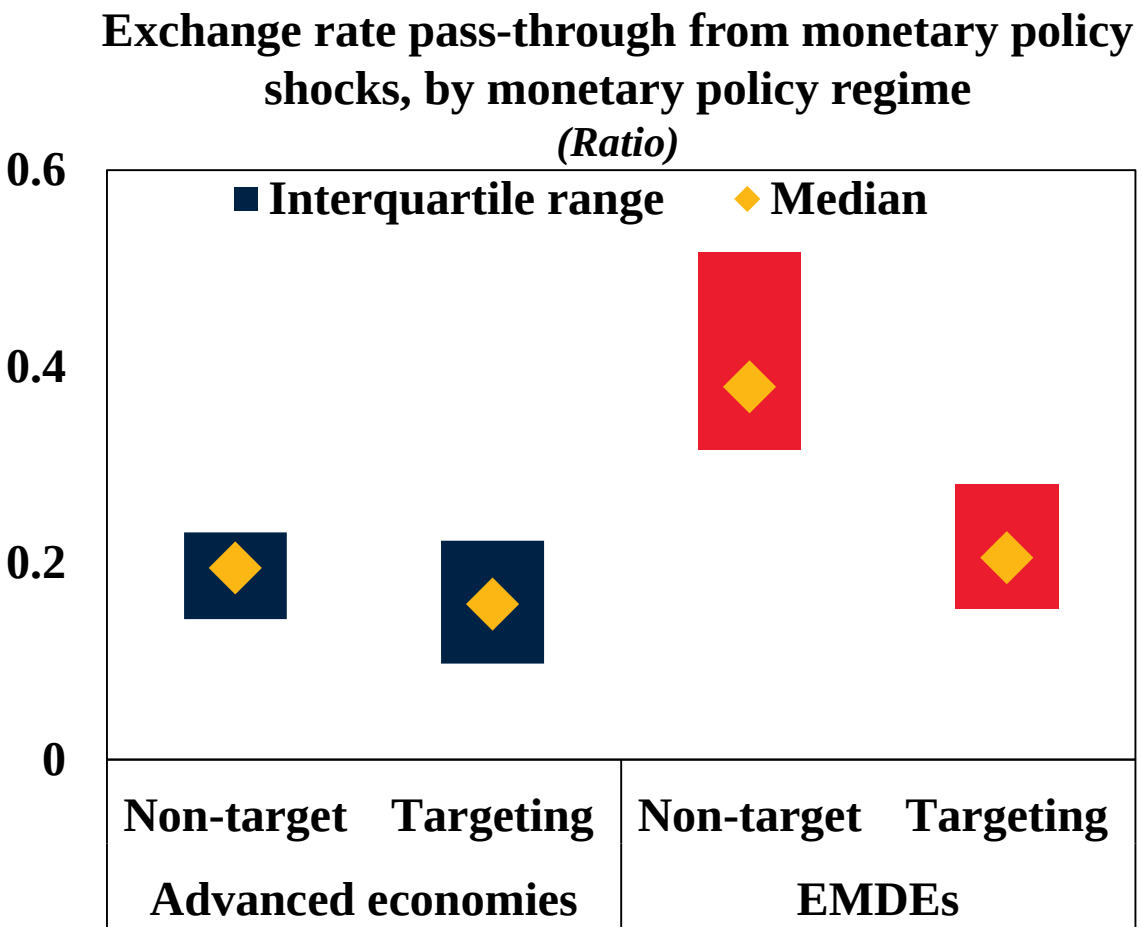
- **Methodology:** Based on FAVAR exercise described earlier

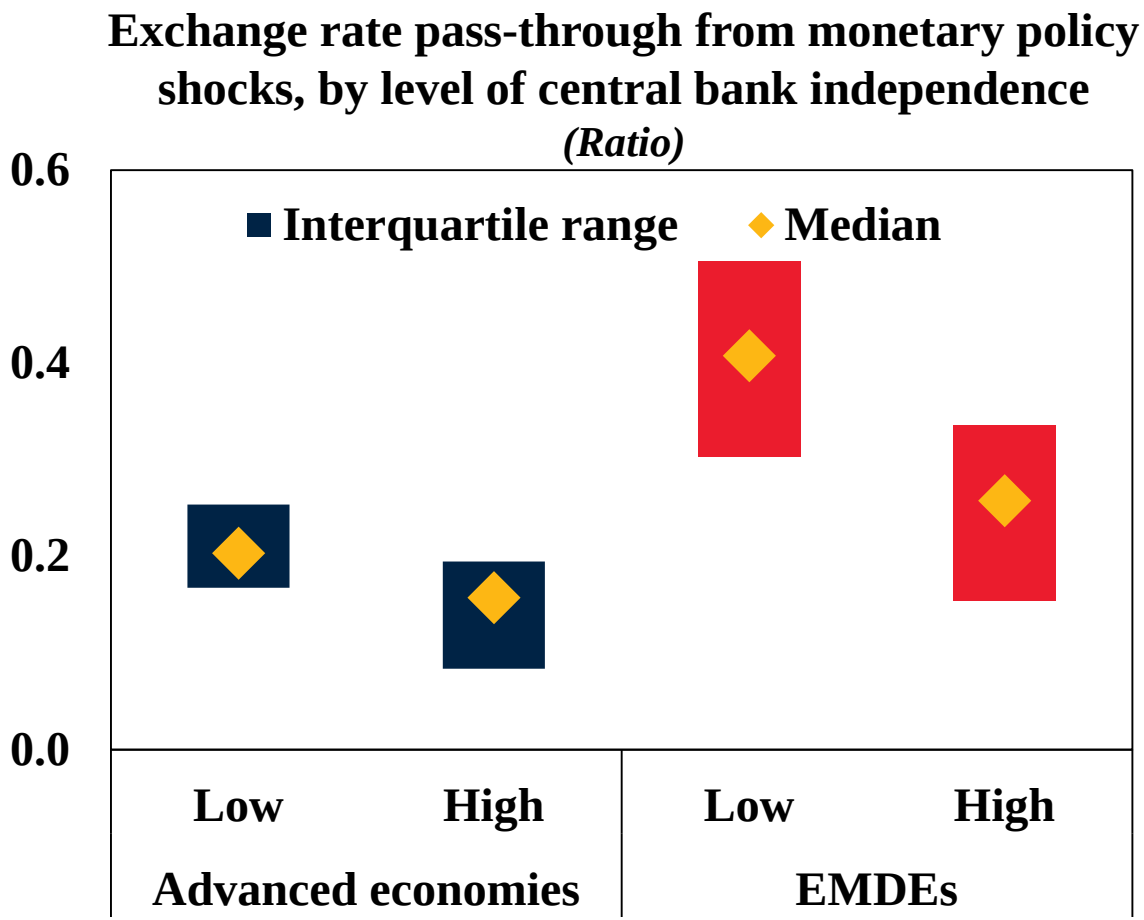
- **Database:** 29 advanced economies and 26 EMDEs for 1970Q1-2017Q4

THE WORLD BANK
IBRD • IDA | WORLD BANK GROUP
Development Economics • Prospects

Exchange Rate Pass-Through - 2

Lower with Inflation Targeting Regime and with Higher Central Bank Independence





Sources: Caceres, Carrière-Swallow, and Gruss (2016); Dincer and Eichengreen (2014); International Monetary Fund; World Bank.

Note: Exchange rate pass-through is defined as the ratio between the one-year cumulative impulse response of consumer price inflation and the one-year cumulative impulse response of the exchange rate change estimated from the factor-augmented vector autoregressive (FAVAR) models for 29 advanced economies and 26 EMDEs over the period 1998-2017. A positive pass-through means that a currency depreciation is associated with an increase in inflation. Diamond markets represent median across countries and bars show the interquartile range. Left Panel. Inflation targeting regimes are defined as in Caceres, Carrière-Swallow, and Gruss (2016) and the IMF Annual Report on Exchange Arrangements and Exchange Restrictions. Right Panel. Central bank independence is measured by the index of central bank independence and transparency in Dincer and Eichengreen (2014). The index ranges from 0 (least independent and transparent) to 15 (most independent and transparent). "Low" and "high" central bank independence are, respectively, defined as below and above the sample average.

THE WORLD BANK
IBRD • IDA | WORLD BANK GROUP
Development Economics • Prospects

Four Questions

1 **How has inflation evolved in EMDEs?** *Declined over time thanks to the sharp fall in global inflation, supported by cyclical and structural developments. If momentum for these developments wanes, maintaining low inflation may become as difficult as achieving it.*

2 **What have been the global and domestic drivers of inflation?** *A wide range of shocks… A global inflation cycle has emerged and inflation synchronization has strengthened among EMDEs over time. Global shocks have become more important over time but domestic shocks remain the main source of national inflation variation.*

3 **How well-anchored are EMDE inflation expectations?** *Despite becoming better-anchored since the 1990s, inflation expectations remain less well anchored in EMDEs than advanced economies. Stronger monetary frameworks and central bank independence are associated with better anchoring of expectations and lower exchange rate pass-through.*

4 **What are the main policy implications?** *Strong monetary policy frameworks, more central bank independence, robust exchange rate regimes and resilient fiscal policy frameworks are all necessary to achieve and maintain low and stable inflation.*

Policy Implications

Learning to Live with the Global Inflation Cycle

1 Establish more resilient monetary policy frameworks

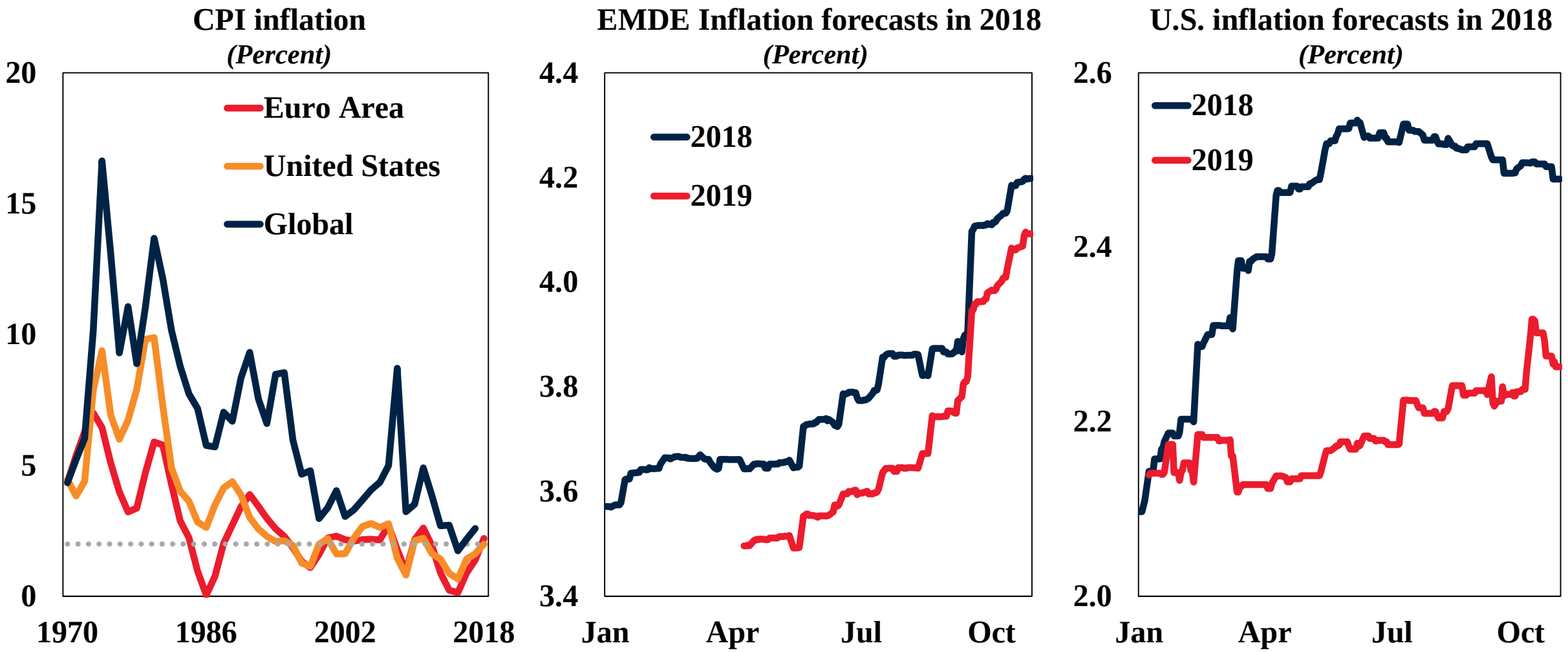
- Greater central bank transparency and independence
- Credible monetary policy frameworks, including inflation targeting
- Robust exchange rate regime

2 Build resilience to changes in global inflation

- Active use of countercyclical policies
- Establishing a fiscal environment resilient enough to effectively contribute to macroeconomic stabilization
- Ensuring stability of financial system
- Considering coordinated monetary policy action to respond to undesirably low or high global inflation

Low Inflation in EMDEs: Miracle or Mirage? - 1

Rising Advanced-Economy Inflation amid Trade Tensions

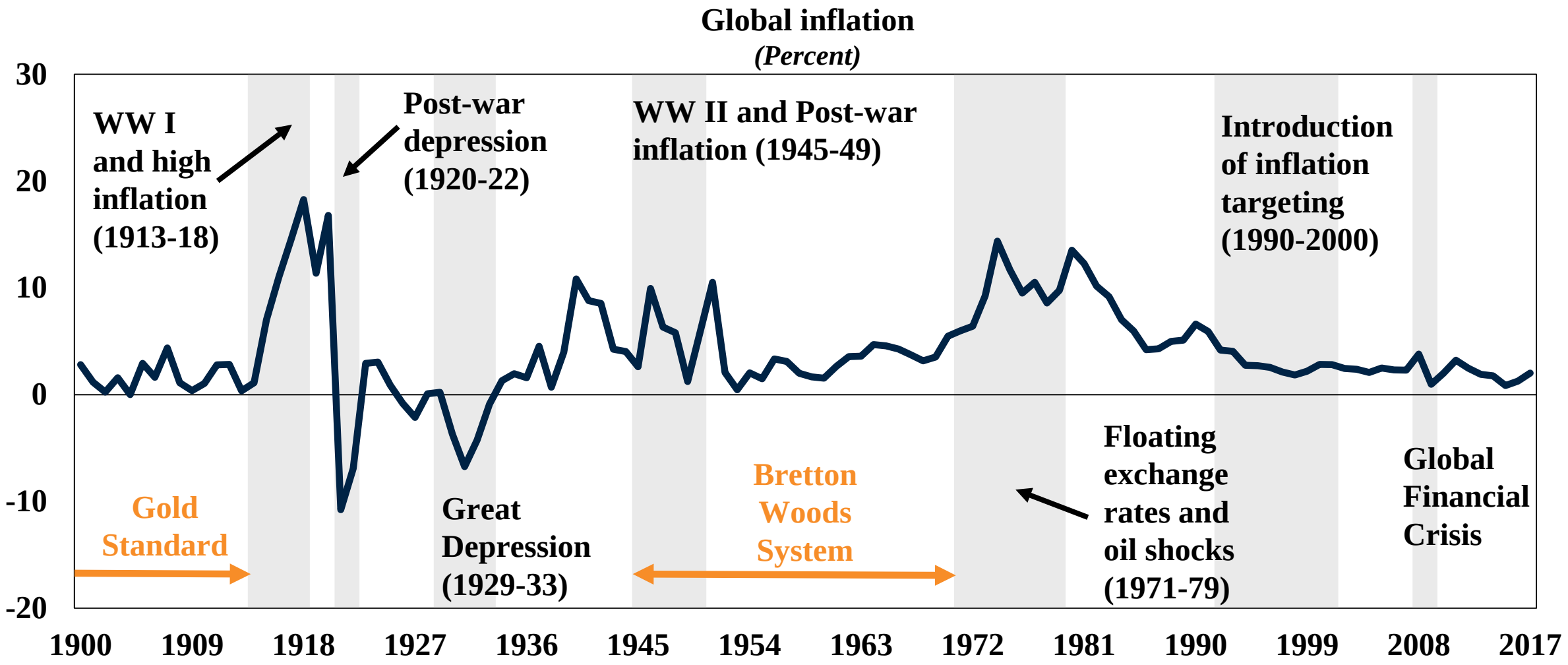


Sources: Consensus Economics, Haver Analytics, World Bank.
Left Panel. Global inflation defined as median of 152 countries. Dotted grey line indicates 2 percent. Center and Right Panels. Consensus forecasts of inflation, as surveyed in respective days of 2018 on the horizontal axis. Last observation is October 29, 2018. Center Panel. Average inflation forecast based on 27 EMDEs for which data of consensus forecasts are available, weighted by current GDP in U.S. dollars for 2018.

THE WORLD BANK
IBRD • IDA | WORLD BANK GROUP
Development Economics • Prospects

Low Inflation in EMDEs: Miracle or Mirage? - 2

Past Periods of Low Inflation Did Not Last



Source: World Bank.
Note: Median of annual average inflation and inflation volatility in 24 countries where data are available across the full period.

THE WORLD BANK
IBRD • IDA | WORLD BANK GROUP
Development Economics • Prospects

Thank You!

Questions & Comments

www.worldbank.org/inflation

THE WORLD BANK
IBRD • IDA | WORLD BANK GROUP
Development Economics • Prospects

Select Publications by Development Prospects Group

- *Global Economic Prospects – January 2019*

  (January and June)

- *Commodity Markets Outlook – October 2018*

  (April and October)

- *Global Monthly*

- *Corporate Debt Overhang and Investment: Firm-Level Evidence – August 2018*
- *The Role of Major Emerging Markets in Global Commodity Demand – June 2018*
- *The 2014–16 Oil Price Collapse in Retrospect: Sources and Implications – April 2018*
- *The Global Costs of Protectionism – December 2017*